# Contents



**Note.** Proofs for many theorems from the course can be found levels 4 or 6 on this website (which have the function of being A level proof documents, those documents can act as a first course in Analysis with a focus on proving stuff used in A levels, and this is a second course in Analysis that will be very useful for later tripos courses).

Note that if you're here for my notes for the tripos course, Analysis I is covered in chapters 1-5 and Analysis II is covered in chapters 6-10, and they have been merged into here.

# 1 Sequences and series

## 1.1 Review and properties of sequences

**Definition.** A sequence is an ennumerated list of real numbers. We can have sequences on any set not just $\mathbb{R}$.

We have learned a million times now (Numbers and sets, level 4, level 6) what it means for a sequence to converge to a value.

Note that when we use an epsilon threshold we can replace it by any non-zero positive multiple of $\varepsilon$ and it won't make a difference.

**Example.** $x_n = \frac{1}{n}$ converges to 0 (See numbers and sets)

$x_n = \frac{1}{2^n}$ also converges to 0 (as for all terms $0 < \frac{1}{2^n} < \frac{1}{n}$, alternatively set $N = \max\left(1, 1 + \log_2\left(\frac{1}{\varepsilon}\right)\right)$ so we know that for all $\varepsilon$ we will be within $\varepsilon$ of 0 after N terms.)

$x_n = i * n$ diverges to infinity, for any threshold L it clears the circle of radius L centered at the origin after (any integer greater than L which exists, see numbers and sets) terms.

$x_n = \sin(n)$ oscillates around and does not converge or go to infinity

Same for $x_n = (-1)^n$

We now have some propositions that are obvious enough that we have assumed them to be true in the past.

**Proposition.** Limits are unique so we can talk about "the" limit.

*Proof.* Suppose $x_n \to L_1$ and $x_n \to L_2$ for a contradiction, then $x_n$ cannot simaltaneously get to and stay within $\left|\frac{L_1+L_2}{2}\right|$ of both limits, so for any epsilon less than $\left|\frac{L_1+L_2}{2}\right|$ it does not work unless $L_1 = L_2$. In particular, we use the triangle inequality: $|L_1 - L_2| \leq |L_1 - x_n| + |x_n - L_2| \leq 2\varepsilon$, and since this is true for every $\varepsilon$ no matter how small we indeed must have $L_1 = L_2$.

□

Imagine this as a picture: A sequence cannot get to and stay within two different disjoint bands.

**Proposition.** Let $x_n,\ y_n$ be real sequences, then if $x_n \leq y_n$ for every n and we have $x_n \to x$ and $y_n \to y$ then $x \leq y$. In particular, if $x_n \leq y_n \leq z_n$ then $x \leq y \leq z$ if $z_n \to z$. This is sort of not obvious because if we replace this theorem with all strict inequalities it is false (You can construct a sequence strictly smaller than another but they converge to the same limit).

*Proof.* We just need to show that y is not smaller than x, which is true because $y_n - x_n$ has all non-negative terms so if it converged to -L then for $\varepsilon < \frac{L}{2}$ we have a problem.

□

**Proposition.** Convergence of complex sequences is equivalent to convergence of their real and imaginary parts.

*Proof.* If the complex sequence converges then $|z| = \sqrt{\mathrm{Re}\,(z)^2 + \mathrm{Im}\,(z)^2} < \varepsilon$ so $|\mathrm{Re}\,(z)| < \varepsilon$. Also, if the real and imaginary parts are less than $\varepsilon$ then

$$|z| = \sqrt{\mathrm{Re}\,(z)^2 + \mathrm{Im}\,(z)^2} \leq \sqrt{2\varepsilon^2} = \sqrt{2}\varepsilon$$

□

**Proposition.** If $x_n \to x$ and $y_n \to y$ then $x_n + y_n \to x + y$

*Proof.* Eventually $|x_n - x| < \varepsilon,\ |y_n - y| < \varepsilon$ after $\max\left(N_x(\varepsilon), N_y(\varepsilon)\right)$ terms, so by the triangle inequality, $|x_n + y_n - x - y| < 2\varepsilon$.

□

**Proposition.** If $x_n \to x$ and $y_n \to y$ then $x_n y_n \to xy$

*Proof.* Eventually,

$$|x_n y_n - xy| \leq |x_n y_n - x_n y| + |x_n y - xy| \leq x_n \varepsilon + y\varepsilon = \varepsilon(x_n + y)$$

however we need to show that $x_n$ has some bound M. This is true because eventually $x_n < x + \varepsilon$ after, say, N terms so it is eventually bounded so it is bounded by $\max(x_1, x_2, \ldots, x_N, x + \varepsilon)$. Bounded meaning there is an M such that all terms are less than M in absolute value.

□

**Proposition.** If $x_n \to x$ and $x_n$ is never 0 and $x$ is not 0 then $\frac{1}{x_n} \to \frac{1}{x}$

*Proof.*

$$\left|\frac{1}{x_n} - \frac{1}{x}\right| = \left|\frac{x - x_n}{x x_n}\right|$$

Evenetually $|x x_n| \geq \frac{|x|^2}{2}$ (take any $\varepsilon < \frac{|x|}{2}$) and $|x x_n| < \varepsilon$ so $\left|\frac{1}{x_n} - \frac{1}{x}\right| \leq \frac{2\varepsilon}{x^2}$, but x is constant so done.

□

**Definition.** Any sequence which is increasing or decreasing is called monotone

**Proposition.** Monotone bounded sequences are convergent

*Proof.* We proved this in numbers and sets. The proof is very simple, just consider the supremum of the sequence (or the infimum if it is decreasing).

□

**Proposition.** Every sequence (not necessarily bounded) has a monotone subsequence

*Proof.* This holds with the same proof as the Bolzano-Weierstrass proof given in level 4. The idea was to consider peaks (values which are never exceeded again), and note that there are either infinitely many (in which case we're done) or an infinitely non-increasing sequence is forced to exist.

□

**Definition.** Let $n_1 < n_2 < \ldots$ be a sequence of natural numbers, then $x_{n_k}$ is a subsequence of $x_n$.

**Proposition.** Every subsequence of a convergent sequence is convergent with the same limit as the main sequence

*Proof.* If the main limit is L, then since $n_k > k$ as n is an increasing sequence of natural numbers, then for each $\varepsilon$ if there is an N such that the main sequence stays within $\varepsilon$ of L, then after that for all k greater than that N, $x_{n_k}$ is $x_j$ for some $j > N$ so it is also within $\varepsilon$ of L.

□

**Proposition.** (nested interval property) Let $I_n$ be a sequence of closed intervals in $\mathbb{R}$ that are each contained within the previous one. Then set $I_n = [a_n, b_n]$. Then if $a_n - b_n \to 0$ then the intersection of all $I_n$'s will be exactly one point.

*Proof.* Since $a_n$ is a monotone sequence bounded above by $b_1$(by nestedness) it converges to some a that is its least upper bound. And $b_n \to b$. But then a=b by additive properties of limits and $a_n - b_n \to 0$.

For all n, $b_n \geq a \geq a_n$ and so the interval contains a, thus the intersection of all these intervals contains a. It contains no other points because, say, if it contained $a + \varepsilon$ for some non-zero epsilon then by definition of convergence we eventually have $b_n \leq a + \frac{\varepsilon}{2}$ which is a contradiction.

□

We will now prove the Bolzano Weierstrass theorem, which says that every bounded sequence has a convergent subsequence. We proved it in level 4 but we will prove it in a different way here.

*Proof.* Let M be a bound for a bounded sequence. Let $I_1 = [-M,\ M]$, then this contains every point of our sequence, so $a_1 = -M,\ b_1 = M$. Now take $c = \frac{a_1+b_1}{2} = 0$. Then if I split $I_1$ into $[a_1, c]$, $[c,\ b_1]$ at least one of these halves will have infinitely many terms, and we will call this half $I_2$. Do this repeatedly to generate a sequence of nested intervals $I_n$ which has a unique common point by the previous lemma.

Construct your sequence by always picking the k'th element of the sequence that is contained in $I_k$ to ensure it is a subsequence (each one will be strctly later in the main sequence than the previous one: the k+1'th one in the k+1'th band occurs at at least the k+1'th position in the k'th band), then it converges to this unique common point. For example, $(-1)^n$ is bounded and not convergent but the subsequence with every other term is convergent.

□

Note that we just know we have at least one convergent subsequence, it is never unique as we can always take a further subsequence.

## 1.2 Cauchy sequences

**Definition:** A Cauchy sequence is a sequence where the elements are getting closer to eachother. I.e. for each $\varepsilon$ there exists an N such that for all $m, n{>}N$, $|a_n - a_m| < \varepsilon$.

Ok, for real numbers these obviously converge in the real numbers (since we go into an arbitrarily small band which means convergence) but we will prove that it is in fact equivalent to convergence in $\mathbb{R}$ (Note that in $\mathbb{Q}$ it fails! The problem is $\mathbb{Q}$ has gaps. It helps to think of a Cauchy sequence that "wants" to converge but we don't know if the "limit point" exists.)

**Example.** $a_n = \frac{1}{n}$ is Cauchy because for any $\varepsilon$ then for n large enough $\frac{1}{n} - \frac{1}{m} < \frac{1}{n} < \varepsilon$.

$a_n = (-1)^n$ is not Cauchy because I can always force the difference to be 2.

Define a sequence by pi like $(3, 3.1, 3.14, 3.141, 3.1415, \ldots)$, then this is cauchy in $\mathbb{Q}$ but does not converge in $\mathbb{Q}$, it only converges in $\mathbb{R}$.

**Proposition.** If a sequence is Cauchy then it is bounded

*Proof.* Fix $\varepsilon = 1$, then there is an N such that for all m>N, $|a_m - a_N| < 1$, so the sequence is eventually bounded by something independent of m and therefore bounded.

□

**Proposition.** A complex sequence is Cauchy if and only if its real and imaginary parts are Cauchy

*Proof.* For any $\varepsilon$ there is an N such that for $m{>}n{>}N$ we have

$$|\mathrm{Im}\,(x_m) - \mathrm{Im}\,(x_n)| < \frac{\varepsilon}{\sqrt{2}}, |\mathrm{Re}\,(x_m) - \mathrm{Re}\,(x_n)| < \frac{\varepsilon}{\sqrt{2}}$$

and therefore

$$|x_m - x_n| = \sqrt{|\mathrm{Im}\,(x_m) - \mathrm{Im}\,(x_n)|^2 + |\mathrm{Re}\,(x_m) - \mathrm{Re}\,(x_n)|^2} < \varepsilon$$

□

**Proposition.** If a sequence converges it is Cauchy

*Proof.* For each $\varepsilon$ there is N depending on $\varepsilon$ such that for $n, m > N$

$$|x_n - x_m| \leq |x_n - x| + |x - x_m| < 2\varepsilon$$

by convergence so it is cauchy.

□

**Proposition.** Real cauchy sequences converge (And by a previous proposition this implies that so do complex cauchy sequences)

*Proof.* Find a subsequence $x_{n_k}$ which converges by the Bolzano-Weierstrass theorem since the sequence is bounded. Let x be the limit of that subsequence. Fix $\varepsilon \to 0$. Then let k be so large that for $m > n \geq n_k$ we have that $|x_n - x_m| < \varepsilon$ and also $|x_{n_k} - x| < \varepsilon$. Now for each m that is that large, by the triangle inequality, setting $n = n_k$, $|x - x_m| \leq |x_n - x_m| + |x - x_n| < 2\varepsilon$.

□

## 1.3 Series and convergence tests

**Definition.** Define an infinite sum series as the limit of its partial sums when it exists, ie $\sum_{n=1}^{\infty} a_n = \lim_{k\to\infty} \sum_{n=1}^{k} a_n$. The sum converges if this limit exists.

**Example.** $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges because

$$\sum_{n=1}^{k} \frac{1}{n\,(n+1)} = \sum_{n=1}^{k} \frac{1}{n} - \frac{1}{n+1} = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{k} - \frac{1}{k+1} = 1 - \frac{1}{k+1} \to 1$$

So the series converges to 1.

**Example.** $\sum_{n=1}^{\infty} n$ clearly does not converge, it definitely does not converge to $-\frac{1}{12}$.

Now we will do some totally trivial and obvious propositions. In fact we have seen these in level 4, so this is just to refresh your memory.

**Proposition.** Fix some complex numbers A and B. If I have two series $\sum a_n, \sum b_n$ that converge then we have that $\sum (Aa_n + Bb_n)$ converges

*Proof.* This is directly from basic properties of convergent sequences.

□

**Proposition.** If two sums agree after some point then they either both converge or neither converge.

*Proof.* Eventually the partial sums will just differ by a constant so it is clear their convergence behavior will be the same.

□

**Proposition.** A sum converges only if the terms approach 0, but this is not a sufficient condition (eg the harmonic series $\sum \frac{1}{n}$ diverges).

*Proof.* If the terms do not approach 0 there is an $\varepsilon$>0 such that the terms are larger in absolute value than $\varepsilon$ for infinitely many n. But $|a_n| > \varepsilon$ implies that $|s_n - s_{n-1}| > \varepsilon$, where $s_n$ is the sequence of partial sums. This implies that $s_n$ is not cauchy since that happens for infinitely many n, completing the proof.

□

**Proposition.** Suppose all terms in a series are positive, then if they are bounded above by a convergent series that series is convergent.

*Proof.* The partial sums are bounded above by the sum of the convergent series in question so the result follows from the monotone convergence theorem (In this course when I say MCT I mean the sequence one not the integral one).

□

I'm pretty sure we did all this in numbers and sets. I remember we used this to show $\sum \frac{1}{n^2}$ converges and now we're doing it again for some reason.

**Proposition.** (Root test) If we are summing non-negative terms $a_n$, then look at $\sqrt[n]{a_n}$ and suppose it converges to a limit a. Then if a<1 the series converges and if a>1 the series diverges. The idea is we are comparing it to a geometric series. We can't say much if a=1.

*Proof.* If $a$>1 then there is an N such that $a_n^{\frac{1}{n}} > 1$ whenever n>N. This implies that $a_n > 1$ and the sum of this clearly diverges.

If $a$<1 fix an r such that $a$<$r$<1, then eventually $\sqrt[n]{a_n} < r$, so we can eventually bound $a_n$ by a geometric series with common ratio r, so apply the previous proposition.

□

**Example.** For $\sum 2^{-n}$ this confirms convergence. For $\sum 4^n$ this confirms divergence. a=1/2 and a=4 respectively in these cases. For $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$ the test is inconclusive.

**Proposition.** (Ratio test) If $\frac{a_{n+1}}{a_n} \to a$ as $n \to \infty$ this converges if $|a| < 1$ and diverges if $|a| > 1$

*Proof.* See level 4 for the proof of this as well as the proof that the above test works with absolute values and non-positive terms.

□

**Example.** $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges as the ratio between consecutive terms approaches $\frac{1}{2}$. In fact (fun exercise) it converges to 2.

**Proposition:** If $f(x)$ is a decreasing non-negative function then $\sum f(n)$ converges if and only if $\int_1^{\infty} f(x)\, dx$ exists and is finite

*Proof.* The reason why this proposition is true is by looking at areas by looking at rectangles and noting that the difference (area of the green rectangles in the image below) is bounded above by $f(1)$ since the function is decreasing. Figure 1 shows why the proposition must be true.

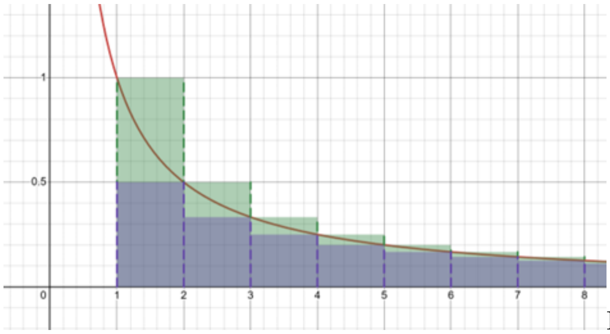

Figure 1

If the sum converges then the sum of the purples converge. The sum of the greens converge by monotone convergence theorem so the integral converges. Conversely if the integral converges subtract the sum of the green to get the sum of the purple which therefore also converges.

□

**Example.** By the integral test we have that $\sum \frac{1}{n^s}$ converges if and only if s>1.

**Example.** $\sum \frac{1}{n\log(n)}$ diverges because its integral is $\log(\log(n))$ which goes to infinity (Very!) slowly.

**Proposition.** (Cauchy condensation test) Let $a_n$ be a decreasing non-negative sequence. Then $\sum a_n$ converges if and only if $\sum 2^n a_{2^n}$ converges.

*Proof.* Since the sum is decreasing we have the following.

$$a_1 + a_2 + (a_3 + a_4) + (a_5 + a_6 + a_7 + a_8) + \cdots \geq a_1 + a_2 + 2a_4 + 4a_8 + \dots$$

So if the left hand side is finite so is the right hand side.

On the other hand

$$a_1 + (a_2 + a_3) + (a_4 + a_5 + a_6 + a_7) + \cdots \leq a_1 + 2a_2 + 4a_4 + 8a_8 + \dots$$

In the end it is clear that both are finite if the other one is finite so that completes the proof.

You can use this to show that $\sum \frac{1}{n\log^2(n)}$ converges, for example.

□

**Proposition.** (Alternating series test) Suppose that $a_n$ is decreasing and tends to 0, then $\sum (-1)^n a_n$ converges. For example, $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \dots$ converges. In fact this converges to $ln(2)$ by Taylor series and Abel's theorem which is proven later in these notes.

Ok, this is obviously true by this figure 2

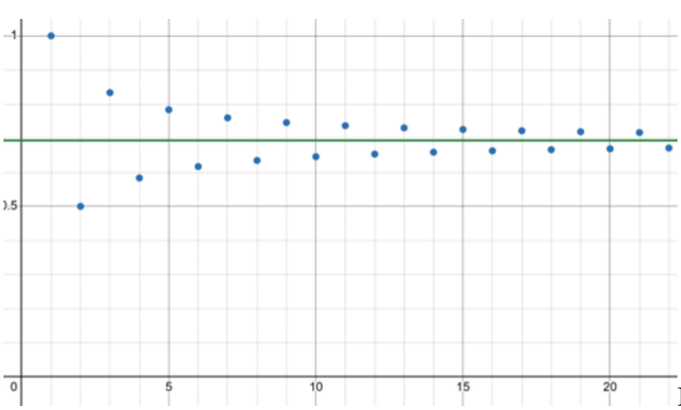

Figure 2

Now lets prove it.

*Proof.* Define $S_n = \sum_{r=1}^{n} (-1)^{1+r} a_r$

Note that $S_{2n} = (a_1 - a_2) + (a_3 - a_4) + (a_5 - a_6) + \cdots + (a_{2n-1} - a_{2n})$ is increasing. Similarly

$S_{2n+1} = a_1 - (a_2 - a_3) - (a_4 - a_5) - (a_6 - a_7) - \cdots - (a_{2n} - a_{2n+1})$ is decreasing.

These are monotone and we show they are bounded as follows. $S_{2n+1}$ is bounded above by $a_1$ and $S_{2n}$ is bounded above by $S_{2n+1}$ since the 2n+1'th term is positive and thus it is bounded above by $a_1$. Similarly both are bounded below by 0. Hence it is clear they both converge by the monotone convergence theorem and that they converge to the same limit since their differences converge to 0.

□

**Proposition.** (Dirichlet test) Let $b_n$ be decreasing and let $a_n$ be a sequence such that its partial sums make a bounded sequence, then the series $\sum a_n b_n$ converges.

*Proof.* Let $S_n = \sum_{j=1}^{n} a_j$ and we say $S_0 = 0$. Now we do some algebra trickery:

$$\sum_{j=m}^{n} a_j b_j = b_n (s_n - s_{n-1}) + b_{n-1} (s_{n-1} - s_{n-2}) + \cdots + b_m (s_m - s_{m-1})$$

$$= s_n b_n - s_{m-1} b_m - s_m b_{m+1} + s_{m+1} b_{m+1} - s_{m+1} b_{m+2} + \cdots + s_{n-1} b_{n-1} - s_{n-1} b_n$$

$$= s_n b_n - s_{m-1} b_m + \sum_{j=m}^{n-1} s_j (b_j - b_{j-1})$$

Now since s is bounded, we can safely write the following. Note that it is not always valid to split lim(a+b) into lim(a)+lim(b) as the left may exist but the right may be, for example, infinity plus minus infinity, but in this case by boundedness we are fine as the non-sum limits exist. We apply the above identity to m=1:

$$\lim_{n\to\infty} \sum_{j=1}^{n} a_j b_j = \lim_{n\to\infty} s_n b_n + \lim_{n\to\infty} \sum_{j=1}^{n} s_j (b_j - b_{j+1})$$

Now

$$\sum_{j=1}^{n} |s_j| \, | (b_j - b_{j+1}) \, | \leq M \sum_{j=1}^{n} |(b_j - b_{j+1})| = M \sum_{j=1}^{n} (b_j - b_{j+1}) = M b_1$$

By the method of differences/telescoping sum, where M is a bound for s, and so the sequence is absolutely convergent. Then since $\lim_{n\to\infty} s_n b_n$ is 0 it follows that $\lim_{n\to\infty} \sum_{j=1}^{n} a_j b_j$ exists as required.

□

A sum $\sum a_n$ is said to be absolutely convergent if $\sum |a_n|$ converges.

We proved this in level 4. I will sketch how the proof went.

*Sketch of proof.* We let $b_n$ be a rearranging of the $a_n$'s and let N be a threshold so that $\sum_{>N} |a_n| < \varepsilon$ and M be so large so that $b_{1\ldots M}$ contains $a_{1\ldots N}$, then it follows that $\sum_{n=1}^{M} b_n$ is within $\varepsilon$ of $\sum a_n$ by the infinite sum variant of the triangle inequality.

**Proposition.** Absolute convergence implies convergence.

*Proof.* Let $s_k = \sum_{n=1}^{k} a_n$ and $r_k = \sum_{n=1}^{k} |a_n|$. The $r_k$'s converge and are thus Cauchy, so for any N and M large enough we have $\sum_{n=N}^{M} |a_n| < \varepsilon$ for each fixed $\varepsilon$. Therefore by the triangle inequality $\left|\sum_{n=N}^{M} a_n\right| < \varepsilon$ for large enough N and M, so the s sequence is Cauchy and hence convergent. A general principle based on this is that you can use the triangle inequality for infinite sums.

□

**Definition.** We say a sequence that is not absolutely convergent is conditionally convergent.

Oh I just realized the example where rearranging terms fails in level 4 + the proof it works for absolute convergent series combined means that the $1 - \frac{1}{2} + \frac{1}{3} + ...$ is conditionally convergent which gives another proof that the harmonic series diverges so now we know 3 proofs (Log bound, powers of 2 proof, and this).

We can say something even stronger: If a series of real numbers is conditionally convergent then there is a rearrangement that can make it converge to *any* value. The idea is that such a series would have its positive and negative terms each add to infinity, so we could use these terms to make it oscillate around any value and eventually get and stay as close as we want to that value.

# 2 Continuous functions

We have discussed in levels 3-7 properties of limits of functions and continuous functions

## 2.1 Basic properties.

**Definition.** If we have a subset of the complex numbers then an isolated point is a point in this set with no other points in the set in some neighbourhood around it. An accumulation point is a point that is not an isolated point, or is a limit of sequence of points in the set, such as the boundary of the unit circle being accumulation points of its interior. Equivalently, if for any $\delta$ there is a point in the set within $\delta$ of that point it is an accumulation point.

Recall that if $f{:}\mathbb{C} \to \mathbb{C}$ then $f$ is continuous at a point $x$ in the domain if for all $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$, and that we also say that if this holds then $f(y)$ tends to $f(x)$ as $y$ tends to $x$, since if there is a sequence tending to $x$ and this holds then f(that sequence) tends to $f(x)$. Similarly the sequential statement does not hold if $f(y)$ does not approach $f(x)$ as y approaches $x$, so these two ideas are equivalent.

We say it goes to infinity if its reciprocal goes to 0, but now we are talking about accumulation points of the domain since a function cannot be infinity in the domain.

Note that a function is trivially continuous at an isolated point in the domain since $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$ is trivial for small enough $\delta$ – by isolation if the ball of radius $\delta$ around x only contains x then $|f(x) - f(y)|$ is automatically 0 and therefore $< \varepsilon$.

Now using our theorems about sequence limits we have that sums or products of continuous functions are continuous, as well as quotients when we are not dividing by 0 near the point in question. It therefore follows that polynomials are continuous since $f(x) = x$ is clearly continuous.

Note that $f(z) \to y$ as $z \to a$ if and only if for every sequence $z_n \to a$, $f(z_n) \to y$. The proof is because if there is a sequence such that $z_n \to a$, $f(z_n)$ does not $\to y$, then that means that for some $\varepsilon$ there are points z within any $\delta$ of a such that $f(z)$ fails to be within $\varepsilon$ of y. Also, if $f(z)$ does not $\to y$ as $z \to a$ then there is an $\varepsilon$ such that for every $\delta$ we can find a z so that have $|z - a| < \delta$ but $|f(z) - f(a)| > \varepsilon$. Now pick a sequence $\delta_n \to 0$ and pick such a $z_n$ for each $\delta_n$, now clearly $z_n \to a$ but $f(z_n)$ does not approach $f(a)$. Applying this to the case $f(a) = y$ gives that continuity and sequential continuity (i.e. saying a function is continuous at a if for every sequence $z_n \to a$ we have $f(z_n) \to f(a)$) are equivalent.

**Example.** $\sin\left(\frac{1}{x}\right)$ is continuous everywhere but 0 where it is not continuous because on any neighbourhood about 0 it takes every value between -1 and 1.

**Proposition.** If $g(x)$ is continuous at x and $f(y)$ is continuous at $y = g(x)$ then $f(g(x))$ is continuous at x.

*Proof.* For all $\varepsilon$ we can find a $\eta$ such that

$$|y - g(x)| < \eta \Rightarrow |f(y) - f(g(x))| < \varepsilon$$

and then for this $\eta$ we can find a $\delta$ such that

$$|z - x| < \delta \Longrightarrow |g(z) - g(x)| < \eta$$

which in turn implies that $|f(g(z)) - f(g(x))| < \varepsilon$. We have done this argument in earlier levels but using words like "arbitrarily close" or "sufficiently close" and now we are doing it symbolically.

□

## 2.2 Extreme value theorem

**Theorem.** (Extreme value theorem) Any continuous function defined on a closed and bounded set attains its maximum, so it is actually a maximum and not just a supremum

*Proof.* I proved this in level 6.2. However, to keep this self contained I will sketch the proof.

We need to suppose it is unbounded then construct a sequence of points where the function is $> n$ at $a_n$ and get a contradiction by noting that there is a convergent subsequence, its limit is in the domain by closure, and by continuity $f$ must be infinite at that point.

We show it attains its maximum by finding $a_n$ such that instead of the function being $> n$ it is $> M - \frac{1}{n}$ where M is its supremum.

□

**Proposition.** The image of closed and bounded sets under continuous complex functions is closed and bounded. This actually implies the extreme value theorem as the closed image has the maximum and the maximum is in the image hence attained by definition.

A closed set contains its supremum because otherwise its complement would contain its supremum and by openness contain a ball around it that means something strictly smaller than the supremum is an upper bound for the set which is a contradiction.

*Proof.* The "bounded" part follows from the extreme value theorem.

As a reminder closed (as in complement is open) is equivalent to closed (as in contains its limit points) because if the complement is open then a limit point not contained in the set has to have an open ball of stuff not in the set around it so it is not a limit point, and if all the limit points are in the set then any point not in the set has to have a ball of points not in the set around it, otherwise it would be a limit point.

Now we do the closed part. Take $y_n$ in $f(x)$ and suppose its limit is y, then we want to show that y belongs to $f(x)$. The sequence of pre-images has a convergent subsequence that converges to some $x$, and then by continuity $f(x) = y$.

□

## 2.3 Intermediate value theorem

**Theorem.** If $f : [a, b] \to \mathbb{R}$ is continuous then given y between $f(a)$ and $f(b)$ there exists a c between a and b such that $f(c) = y$. So continuous functions map closed intervals to closed intervals, not just closed sets.

This one is obvious, and in fact we have secretly used before, but it's not too obvious since it is false over $\mathbb{Q}$ for example, but if you have a sense of how the real numbers work then it is indeed obvious.

*Proof.* If $f$ is constant then this is trivially true. If not then we can take $f(a)$ to be the minimum and $f(b)$ to be the maximum since otherwise we could just take the minimum and maximum and work with those points. Set $S = \{x \in [a, b], \ f(x) \leq y\}$. Then a is in S and S is non-empty and S is bounded above by b so S has a supremum d. Now $f(d) \leq y$ because if $f(d) > y$ then $y - f(d)$ is positive, and we can call the difference $\varepsilon$. Now by continuity there is a $\delta$ such that for z within $\delta$ of d $y - f(z)$ is positive. But then this contradicts d being the supremum, because $d - \delta$ is an upper bound.

On the other hand, suppose that $f(d) < y$, then for the same reason, if $y - f(d) = -\varepsilon$ then for some $\delta$ we have that $d + \delta$ is in S so d is not an upper bound for S, which is another contradiction since we supposed d was the supremum so we are done.

□

**Example.** Now we know that $\sqrt{2}$ exists because $x^2$ hits 2 somewhere on $[1, 2]$. Ok but we already knew that $\sqrt{2}$ exists anyway but sometimess we like to be *unnecessarily* pedantic about stuff. In fact all n'th roots and logs and stuff exist.

**Definition.** A function $[a, b] \to \mathbb{R}$ is called monotone if it is increasing or decreasing, meaning that if $x_1 \leq x_2$ then $f(x_1) \leq f(x_2)$. We say a function is strictly increasing if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$.

**Proposition.** Strictly increasing or strictly decreasing continuous functions have continuous strictly monotone inverses.

*Proof.* By the IVT each point has a pre-image and by strictly-monotone-ness it is unique, so there is an inverse. The hard (but kind of obvious if you think about it) part is to show that it is actually continuous and strictly monotone.

Take $f$ to be strictly increasing because we can do a symmetric argument for the other case.

Lets show that its inverse is strictly increasing. If $f^{-1}$ is not strictly increasing then there exist points such that $a < b$ but $f^{-1}(a) \geq f^{-1}(b)$ which contradicts f being strictly increasing. Doing f to both sides gives a contradiction.

Fix $y_0 \in [c, d]$ where we consider f as a function from $[a, b] \to [c, d]$. Then we have 2 cases. Lets say that $x_0$ is the pre-image of $y_0$.

Case 1: $x_0 \in (a, b)$, then for any $\varepsilon$ there is an $\eta$ such that for some $0 < \eta \leq \varepsilon$ we have it small enough that $[x_0 - \eta, x_0 + \eta] \subseteq [a, b]$ by openness.

Now $x_0 - \eta < x_0 < x_0 + \eta$ so $f(x_0 - \eta) < y_0 < f(x_0 + \eta)$ as f is strictly increasing. So if we take $\delta$ to be $\min\{f(x_0 + \eta) - y_0, y_0 - f(x_0 - \eta)\}$ then indeed we get that $|y - y_0| < \delta$ implies that $\left|f^{-1}(y) - x_0\right| < \eta \leq \varepsilon$ so we have continuity so done.

Case 2: $x_0 = a$ or $x_0 = b$. By symmetry, pick $x_0 = a$.

We just apply case 1 to the function that has had its domain extended to the left of a as something like $f(x) = \begin{cases} x + c - a \text{ if } x < a \\ \text{normal } f(x) \text{ otherwise} \end{cases}$ and show that this is continuous at a by step 1 and thus normal f is continuous there too.

□

# 3 Differentiation

## 3.1 Basic properties

Differentiation is the same limit that we already know from A levels, and we recall that it has to exist from the left and from the right for a function to be differentiable there, and from all directions to be complex differentiable – the complex conjugate is not complex differentiable because from the i direction it looks like -x and from the real direction it looks like +x so the limit is -1 from some directions and 1 from others. But we will do more stuff on complex differentiable functions in the complex analysis course.

We cannot define a derivative at an isolated point in a domain, for there to be a limit there has to be points arbitrarily close.

Basic properties like the product rule are easy to derive by messing with the limits - we did this in earlier levels so I leave it as an exercise.

**Proposition.** We have the chain rule, which says

$$\frac{d}{dx}g(f(x)) = g'(f(x))f'(x)$$

*Proof.* We proved this in Level 4. It is easy to give a "proof" using limits that has a divide by 0 issue. The idea to get around this was to mess with the function

$$d(y) := \begin{cases} \frac{g(y)-g(f(x))}{y-f(x)} & y \neq f(x) \\ g'(f(x)) & y = f(x) \end{cases}$$

and split into cases $g\left(f\left(x+h\right)\right) = g\left(f\left(x\right)\right)$ and $g\left(f\left(x+h\right)\right) \neq g\left(f\left(x\right)\right)$

□

Remember that the derivative equivalently says

$$f\left(a+h\right) = f\left(a\right) + f'\left(a\right)h + o\left(h\right)$$

**Example.** $x * \sin\left(\frac{1}{x}\right)$ is differentiable everywhere but 0 where $\lim_{h\to 0}\frac{f(h)}{h} = \lim_{h\to 0}\sin\left(\frac{1}{h}\right)$ which is not a limit that exists.

Here's another obvious one.

**Proposition.** Differentiability implies continuity

*Proof.*

$$\lim_{x\to a} f(x) = \lim_{h\to 0} f\left(a+h\right) = \lim_{h\to 0} f\left(a\right) + hf'\left(a\right) + o(h) = 0$$

□

**Proposition.** Rolle's theorem, ie if $f$ is continuous on $[a,b]$ and differentiable on $(a,b)$ and $f(a) = f(b)$ then $f'(c) = 0$ for some $c \in (a,b)$

*Proof.* We did this in level 6.2, but again I will sketch the proof. If $f$ is constant the theorem is trivial, and otherwise the extreme value theorem asserts the existence of either an interior minimum or maximum, at which point messing with limits implies the derivative is 0.

□

**Theorem.** Mean value theorem, if $f$ is continuous on $[a,b]$ and differentiable on $(a,b)$ then $f'(c) = \frac{f(b)-f(a)}{b-a}$ for some $c \in (a,b)$

*Proof.* Apply rolle's theorem to $g\left(x\right) = f\left(x\right) - \frac{f(b)-f(a)}{b-a}x$

□

A corollary of the mean value theorem is the already known result that if the derivative is non-negative everywhere on an interval the function is increasing on the interval - if it were decreasing the derivative would be negative at some point. We also get that zero derivative on an interval (NOT an arbitrary subset of the real numbers) implies our function is constant.

**Theorem.** (1D inverse function theorem) If we have a function $f : [a, b] \to \mathbb{R}$ continuous and differentiable on $(a, b)$ and $f'(x) > 0$ everywhere on $(a, b)$, then f is a bijection and furthermore f has an inverse which is continuous and differentiable except possibly at its end points.

Later we will prove a generalization of the inverse function theorem to higher dimensions.

We note that the derivative of our inverse function is the reciprocal of the derivative of the original function at the corresponding point – of course, we will show this.

*Proof.* The bijection part and continuous part follows from a previous theorem because f is strictly increasing by a previous proposition. Therefore we just need to prove the differentiability part.

Let $y \in (f(a), f(b))$ and let x be its unique pre-image. Given h such that $y+h$ is still inside the open interval $(f(a), f(b))$ take k to be a number such that $y+h = f(x+k)$ so $k = f^{-1}(y+h) - x$. Now we want to consider $\frac{\left(f^{-1}(y+h)-x\right)}{h}$ because if this approaches a limit as h goes to 0 that is the derivative we want. We get $\frac{k}{f(x+k)-f(x)}$ which as h goes to 0 (and therefore k goes to 0 by continuity) we approach the reciprocal of f'(x), as desired.

□

**Example.** This means that stuff like arcsin or log or roots are differentiable, without being like "obviously just look at it".

**Proposition.** (Cauchy mean value theorem) Let f,g be functions taking real values continuous on [a,b] and differentiable (a,b). Then there exists a c in (a,b) such that

$$g'(c)(f(b) - f(a)) = f'(c)(g(b) - g(a))$$

*Proof.* Try the function

$$h(x) = [g(x) - g(a)][f(b) - f(a)] - [g(b) - g(a)][f(x) - f(a)]$$

Now

$$h'(x) = g'(x)(f(b) - f(a)) - f'(x)(g(b) - g(a))$$

This is 0 at some point because $h(a) = h(b)$ and Rolle's theorem is true.

□

Now we will state L'hopital's rule

Suppose that the following hold:

1. $f(a) = g(a) = 0$
2. $g(x)$ is not 0 if $x$ is within some neighbourhood of c
3. $f$ and $g$ are differentiable in the vicinity of $a$ and $\lim_{x \to c} \frac{f'(x)}{g'(x)}$ exists and is equal to $l$

Then $\frac{f(x)}{g(x)} \to l$ as $x \to a$.

In the lecture, the proof of this was left as an exercise. However, the proof was done in level 6.2.

Note that derivatives don't have to be continuous, for example the derivative of $x^2 \sin\left(\frac{1}{x}\right)$ exists at all non-zero points, is 0 at $x = 0$ and oscillates wildly near $x = 0$ so it is not continuous at 0.

## 3.2 Taylor's theorem

**Definition.** We say a function $f$ is smooth if it is infinitely differentiable.

Recall generalized Rolle's theorem from level 6.2, which says

Let $f$ be continuous on $[a, b]$ and n-1 times continuously differentiable, but also n times differentiable on the open interval $(a, b)$, and suppose also that

$$f(a) = f'(a) = f''(a) = \cdots = f^{(n-1)}(a) = f(b) = 0$$

then there is some $x$ in $(a, b)$ such that $f^{(n)}(x) = 0$.

The proof was an easy induction using normal Rolle's theorem.

**Theorem.** If $f$ on $[a, a+h]$ is continuous and also n-1 times continuously differentiable and on $(a, a+h)$ it is n times differentiable, then the error of the n'th order taylor polynomial about a at a+h by generalized Rolle's theorem $\frac{1}{n!}\,h^n\,f^{(n)}(a+ch)$ for some c between 0 and 1.

*Proof.* Again, since we proved this in Level 6.2, we will sketch the proof. The result follows easily from the Generalized Rolle's theorem, by subtracting a polynomial cleverly to make the conditions hold.

□

However we can also get the Cauchy form of remainder for Taylor's theorem:

*Proof.* We get (if we expand about $a + h$ instead of a) the remainder term is as follows (supposing a=0 as we can shift):

$$g(t) = f(h) - \sum_{k=0}^{n-1} \frac{f^{(k)}(t)}{k!}(h-t)^k$$

Note that $g(h) = 0$

$$g'(t) = -\sum_{k=0}^{n-1} \frac{f^{(k+1)}(t)}{k!}(h-t)^k + \sum_{k=0}^{n-1} \frac{f^{(k)}(t)}{(k-1)!}(h-t)^{k-1}$$

Note that most of these terms cancel so we get

$$-\frac{f^{(n)}(t)}{(n-1)!}(h-t)^{n-1}$$

By product and chain rules.

Now set $\phi_p(t) = g(t) - \left(\frac{h-t}{h}\right)^p g(0)$ so $\phi_p(0) = \phi_p(h) = 0$ and p is one of 1,2,3,. . . ,n.

Now by Rolle's theorem there is some C between 0 and 1 such that

$$0 = \phi_p'(Ch) = g'(Ch) + \frac{p(1-C)^{p-1}}{h} g(0).$$

But then this means that

$$g(0) = -\frac{hg'(Ch)}{p(1-C)^{p-1}} = \frac{h^n}{p(n-1)!}(1-C)^{n-p} f^{(n)}(Ch)$$

where I plugged in the definition of g and simplified.

Since $g(0)$ is the remainder of our taylor series at h, we call this the cauchy form of remainder. This gives us another version of Taylor's theorem alternative from the 2 we had in level 6 technical results.

Now for smooth functions, we know that the error is $O(h^n)$ but we do not know that it goes to 0 as n goes to infinity.

However, if $f^{(n)}$ does not grow faster than n! anywhere on $[a, a+h]$ then the error term $\frac{1}{n!}\ h^n\ f^{(n)}\,(a+ch)$ does go to 0. This is a more general proof that Taylor series work than the direct proof we gave in level 4 that can be adapted to most standard functions. In complex analysis we will see another proof that Taylor series work with a more elegant condition so that will be fun :)

□

**Example.** Suppose we have $x^q$ for q rational but not an integer, which is smooth on $(0, \infty)$.

Now suppose we taylor expand it about $x = 1$, then the remainder after n-1 terms of the polynomial will be of the form

$\left(\begin{array}{c} q \\ n \end{array}\right)(1+tx)^{q-n}\ x^n$ for some t between 0 and 1 since that is the n'th derivative.

Here $\left(\begin{array}{c} q \\ n \end{array}\right)$ simply means $\frac{q(q-1)\ldots(q-n+1)}{n!}$

We deduce the generalized binomial theorem because $\left|\left(\begin{array}{c} q \\ n \end{array}\right)(1+tx)^{q-n}x^n\right| \approx \left|\left(\begin{array}{c} q \\ n \end{array}\right)x^n\right|$ which goes to 0 provided $|x| < 1$.

We can use the cauchy remainder: We have $(1-t)^{n-1}n\left(\begin{array}{c} q \\ n \end{array}\right)(1+tx)^{q-n}x^n$

We can write this as $q\left(\begin{array}{c} q-1 \\ n-1 \end{array}\right)\left(\frac{1-t}{1+tx}\right)^{n-1}(1+tx)^{q-1}h^n$, and we observe that $\left(\frac{1-t}{1+tx}\right)^{n-1} \to 0$ whenever x is between -1 and 1 and t is between 0 and 1, since the stuff in the brackets is less than 1. We have some constants and we are left with $\left(\begin{array}{c} q-1 \\ n-1 \end{array}\right)h^n$ which is just $\left(\begin{array}{c} q-1 \\ n-1 \end{array}\right)$ but we have a term going to 0 exponentially fast so we are fine.

By doing it this way we see that the error gets small exponentially and faster for smaller x.

As usual we say a function is analytic if it locally has a taylor series converging to the function.

# 4 Integration

Recall from level 4 that the Riemann integral is defined if the lower estimates and upper estimates for the area can get within $\varepsilon$ of eachother for every $\varepsilon$, and that this always exists for continuous functions on closed bounded intervals.

However, this is not the standard definition, despite the fact that it is easy to show that this is equivalent to the standard definition, which is as follows.

Let $f$ be any bounded function on $[a, b]$, then $\underline{\int_a^b} f(t)dt$ is the supremum over all partitions of $[a, b]$ of the integrals of sums of rectangles that underestimate the function at each partition. This is called the power integral. We similarly define the upper integral as $\overline{\int_a^b} f(t)dt$. $f$ is said to be Riemann integrable if these integrals are equal.

**Proposition.** This is equivalent to the $\varepsilon$ definition

*Proof.* Fix $\varepsilon > 0$. If $f$ is integrable according to the definition introduced above, then there exist partitions $D_1$ and $D_2$ such that if we denote $U_{D_1}$ and $L_{D_2}$ as the upper and lower integrals with these partitions respectively, then

$$U_{D_1} < \int f(x)dx + \frac{\varepsilon}{2}$$

$$L_{D_2} < \int f(x)dx - \frac{\varepsilon}{2}$$

Let $D$ be any partition that contains the end points of both $D_1$ and $D_2$, then since refining a partition makes you get closer to the actual integral (clearly),

$$U_D - L_D \leq U_{D_1} - L_{D_2} < \varepsilon$$

Conversely, if the above inequality holds, then this implies that $\inf U_D - \sup L_D < \varepsilon$ for every $\varepsilon$ so they are equal, since clearly any upper estimate is larger than any lower estimate so the difference is actually always positive.

$\square$

**Example.** If f is 0 at irrational numbers and 1 at rational numbers it is not riemann integrable because any interval will have some 0 and some 1 so the lower and upper estimates will differ by the size of the interval and will not get arbitrarily small. However this f is Lebesgue integrable.

If we refine a partition we will get a better estimate as if we do a lower estimate then refine the partition then clearly the lower estimate will be at least as good. A refinement formally means that it contains the previous partition.

**Example.** $f(x) = x$ on [0,1] is riemann integrable because if we take a partition like $\left(0, \frac{1}{n}, \frac{2}{n}, \ldots, 1\right)$ then

Lower sum is $\sum_{i=0}^{n-1} \frac{1}{n} * \frac{i}{n}$ and upper sum is $\sum_{i=1}^{n} \frac{1}{n} * \frac{i}{n}$. So we have $\frac{1}{2} \pm \frac{1}{2n}$ as our sums, and so we see that we have an integrable function with integral $\frac{1}{2}$.

The function on [0,1] that is 0 up to and including $\frac{1}{2}$ and then 1 after because for each $\varepsilon$ we can take the partition $\left(0, \frac{1}{2} - \frac{\varepsilon}{2}, \frac{1}{2} + \frac{\varepsilon}{2}, 1\right)$ and the difference between the lower and upper estimates will be less than $\varepsilon$.

**Proposition.** A function is riemann integrable if it is continuous on a closed bounded interval

*Proof.* We proved this in level 4. However, the idea was we could suppose it is false and exhibit such an $\varepsilon$ where it fails and use the Bolzano weierstrass theorem to get that it cannot be continuous at said point a contradiction.

$\square$

**Proposition.** An increasing function on a closed bounded interval is riemann integrable

*Proof.* If we take the partition $\left(0, \frac{1}{n}, \frac{2}{n}, \ldots, 1\right)$ then we have to multiply the gap size by the differences but the sum of the differences is f(b)-f(a) so the total difference is bounded by $\frac{f(b)-f(a)}{n}$ which goes to 0 as n goes to infinity.

$\square$

**Definition.** The mesh of a partition is the largest size of any interval in the partition.

**Proposition.** Let $f$ be riemann integrable on $[a, b]$. Then for all $\varepsilon > 0$ there exists a $\delta > 0$ such that any partition with mesh $< \delta$ has the upper and lower sum $< \varepsilon$.

*Proof.* By integrability of $f$ we can find some partition D such that its upper and lower sum is $< \frac{\varepsilon}{2}$. Suppose that this partition has $N$ intervals and that $f$ is bounded, say always in $[-M, M]$, since boundedness is needed for riemann integrability by definition. Then let P be any partition with mesh $< \frac{\varepsilon}{4NM}$. Then let Q be the common refinement of D and P. Then clearly, $U_Q - L_Q < \frac{\varepsilon}{2}$. We want to consider how much $U - L$ changes when we remove D from Q. Note that we will remove at most $N$ things from D, and each thing we remove will be on some interval in P, and after removal this interval in P will have its $U - L$ be $< 2M\frac{\varepsilon}{4NM} = \frac{\varepsilon}{2N}$. Since there are at most N intervals in P that will be affected by removing D, and each will be affected by at most $\frac{\varepsilon}{2N}$, the whole $U - L$ sum will be affected by at most $\frac{\varepsilon}{2}$. Thus, it follows that $U_P - L_P < \varepsilon$, so we're done since $P$ was arbitrary and $\frac{\varepsilon}{2NM}$ only depends on $\varepsilon$.

$\square$

We know that continuous functions are integrable, and it immediately follows that piecewise continuous functions are integrable by first partitioning into the pieces. However, there is a technical detail.

Note that if f and g are bounded above by M integrable and disagree except on a finite set then the integrals are the same as they can be made as close to eachother as we want by making a partition that isolates the "bad points" with a small enough interval around them in the partition, specifically make the width at most $\frac{\varepsilon}{2Mk}$ where k is the number of bad points, so that the difference between the upper and lower sums in these intervals is at most $\frac{\varepsilon}{k}$ and there are at most k of them so the total difference is at most $\varepsilon$. We now know that piecewise continuous implies integrable (as it no longer matters what f is at the points of discontinuity).

We can formally prove stuff like that sums and constant multiples of integrable functions are integrable, here is how you might do that:

If f and g are integrable take a partition that makes the difference less than $\frac{\varepsilon}{2}$ and take the least common partition and the result follows.

If f is integrable and we want Nf to be integrable take a partition that makes the difference less than $\frac{\varepsilon}{N}$ then the result follows.

Integrable functions can get bad. As an example, consider the following function:

$$f : [0,1] \to \mathbb{R}$$

$$\begin{cases} 0 \text{ if } f \text{ is irrational} \\ \frac{1}{q} \text{ if } f \text{ is } \frac{p}{q} \text{ when fully simplified} \end{cases}$$

Figure 3 shows a graph of this function.

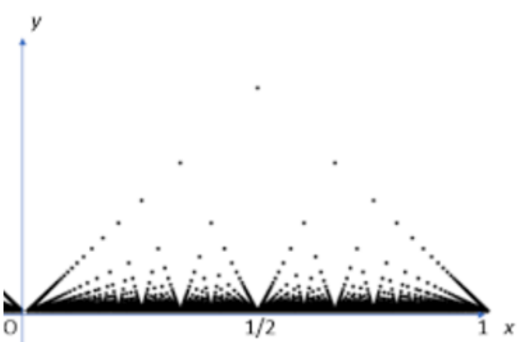

Figure 3

Note that the lower integral is always 0, so we want to show that we can make the upper integral as small as we want by making a clever partition.

Fix $\varepsilon > 0$. Now for all finitely many points that are greater than $\frac{\varepsilon}{2}$ in size, say there are M of them, cover them each by an interval of length $< \frac{\varepsilon}{2M}$. Now in these parts of the partition the upper sum will contribute at most $\frac{\varepsilon}{2}$, but this is also the case in the rest of the parts of the partition, so we have total less than $\varepsilon$.

The underlying reason is that the function is continuous at every point that is not a rational number and that bounded functions on closed bounded intervals are integrable if and only if their discontinuity set has measure 0 (See level 6 for what this means, roughly it means 0 length/area/volume). We proved one of the directions in level 6 and will see the other direction later here.

**Lemma.** For all functions f on any interval I,

$$\sup_I |f| - \inf_I |f| \leq \sup_I f - \inf_I f$$

*Proof.* If f is always positive or always negative this is immediate. Otherwise, if $\inf f < 0 < \sup f$ then we have that

$$\sup_I |f| - \inf_I |f| \leq \sup_I |f| \leq \max\left\{\sup_I (f), \sup_I (-f)\right\} \leq \sup_I (f) + \sup_I (-f) = \sup_I f - \inf_I f$$

□

**Lemma.**

$$\sup_I f^2 - \inf_I f^2 \leq 2\sup_I |f| \left( \sup_I(f) - \inf_I (f) \right)$$

*Proof.*

$$f(x)^2 - f(y)^2 = (f(x) + f(y)) \ (f(x) - f(y)) \leq 2\sup |f| (\sup (f) - \inf (f))$$

So take supremums on both sides.

□

**Proposition.** $\left|\int f\right| \leq \int |f|$ if f is integrable.

*Proof.* Note that by 2 lemmas ago, it follows that if $f$ is integrable, then so is $|f|$. It then follows that since $f$ is between $-|f|$ and $f$, $-\int |f| \leq \int f$ and $\int -f \leq \int |f|$.

Note that this generalizes to complex integration – the idea is to write the integral as $Re^{i\theta}$, but we will do it in the complex analysis course.

□

**Proposition.** If $f$ and $g$ are integrable and bounded then so is $fg$

*Proof.* Write $fg = \frac{1}{4}\left[(f+g)^2 - (f-g)^2\right]$. Then by 2 lemmas ago we have on each part of a partition $U\left(f^2, P\right) - L\left(f^2 - P\right) \leq 2\sup |f| \left[U(f,P) - L(f,P)\right]$. Clearly, f+g and f-g are integrable, so just applying this to $f = f+g$ and $f = f-g$ we get the result.

□

**Proposition.** Let $f : [a,b] \to \mathbb{R}$ be riemann integrable and bounded, and set $F(x) = \int_a^x f(t)\,dt$, then F is continuous on [a,b].

*Proof.* Note that by 2 lemmas ago,

$$|F(x+h) - F(x)| \leq \int_x^{x+h} |f(t)|\,dt \leq h * \sup_{[a,b]} |f|$$

then this goes to 0 as h goes to 0.

□

**Theorem.** (Fundamental theorem of calculus) Let $f : [a,b] \to \mathbb{R}$ be continuous (and therefore riemann integrable and bounded), then F as above has derivative f. This is the fundamental theorem of calculus, which we convinced you of in levels 3 and 4 but now we will fully formalize it.

*Proof.* Set $\varepsilon(h) = \frac{F(x+h) - F(x) - hf(x)}{|h|}$, then we want to show that $\varepsilon(h) \to 0$ as $h \to 0$.

We have a bound for the numerator:

$$|F(x+h) - F(x) - hf(x)| = \left|\int_x^{x+h} f(t)dt - hf(x)\right| = \left|\int_x^{x+h} f(t) - f(x)dt\right|$$

$$\leq \int_x^{x+h} |f(t) - f(x)|\,dt \leq h * \sup_{t\in[0,h]} |f(x+t) - f(x)|$$

and by continuity of f it follows that $\varepsilon\left(h\right)\to0$ as $h\to0$ as required. (Symmetric argument if h is negative)

□

*Remark.* We need continuity otherwise we could have a jump function and integrate it to get something like an absolute value function which is not differentiable everywhere.

We also know that the integral of a derivative is the original function (up to a constant) when the derivative is continuous.

**Theorem.** (Fundamental theorem of calculus again) Let f on [a,b] be riemann integrable and suppose that it has an antiderivative F.

Then $F\left(b\right)-F(a)=\int_{a}^{b}f\left(t\right)dt$

*Proof.* For all $\varepsilon>0$ there exists a partition P of [a,b] such that the upper and lower sums differ by at most $\varepsilon$. By the mean value theorem, for each j, there is a $t_{j}$ between $x_{j-1}$ and $x_{j}$ such that $F\left(x_{j}\right)-F\left(x_{j-1}\right)=f\left(t_{j}\right)\left(x_{j}-x_{j-1}\right)$. But the sum of everything on the left hand side is $F\left(b\right)-F\left(a\right)$ so it follows that $\sum_{j=1}^{n}f\left(t_{j}\right)\left(x_{j}-x_{j-1}\right)=F\left(b\right)-F(a)$. We know that this is somewhere between the lower sum and the upper sum since $t_{j}$ is between the point where f achieves its supremum and infimum. Because this is true for all partitions, the result follows.

□

Note that it is not automatic that a derivative is riemann integrable, but constructing a bounded yet non-integrable derivative is not trivial but can be done with the result that integral implies continuous almost everywhere (level 6.2). The idea is to use a fat cantor set to make the discontinuity set of the derivative have non-zero measure.

As an example, we can now prove that if f and g are continuously differentiable then integration by parts works, i.e.

$$\int_{a}^{b}f'g=f\left(b\right)g\left(b\right)-f\left(a\right)g\left(a\right)-\int_{a}^{b}fg'$$

We know how to prove this because we have derived integration by parts before. It's reverse product rule.

Again, integration by substitution works: If f is continuous and g is continuously differentiable and g(a)=A and g(b)=B then $\int_{a}^{b}f\left(x\right)dx=\int_{A}^{B}f\left(g\left(t\right)\right)g'\left(t\right)dt$. Again the proof is the same – we just use the fundamental theorem of calculus and the chain rule.

**Theorem.** (Taylor's theorem with integral form of remainder)

Let f be n+1 times differentiable on [a,b] with it's (n+1)'th derivative continuous. Then we have that $f\left(b\right)=f\left(a\right)+\left(b-a\right)f'\left(a\right)+\frac{\left(b-a\right)^{2}}{2!}f^{(2)}\left(a\right)+\cdots+\frac{\left(b-a\right)^{n}}{n!}f^{(n)}\left(a\right)+\int_{a}^{b}\frac{\left(b-t\right)^{n}}{n!}f^{(n+1)}\left(t\right)dt$.

*Proof.* As in level 6.2, we can do this by induction/integration by parts.

We can alternatively write the remainder at h after the (n-1)'th derivative term as

$$\frac{1}{(n-1)!}\int_{0}^{h}\left(h-u\right)^{n-1}f^{(n)}\left(a+u\right)dt$$

I write it this way just because that is how it was written in the lecture.

□

**Proposition.** Let $f,g:[a,b]\to\mathbb{R}$ be continuous such that g is never 0. Then there exists $c\in[a,b]$ such that we have $\int_{a}^{b}f\left(x\right)g\left(x\right)dx=f(c)\int_{a}^{b}g\left(x\right)dx$

*Proof.* Apply the cauchy mean value theorem to $F(x) = \int_a^x f(t)\, g(t) dt,\ G(x) = \int_a^x g(t)\, dt$. We get

$$g(c) \int_a^b f(x)\, g(x)\, dx = [F(b) - F(a)]\, G'(c) = F'(c)\, [G(b) - G(a)] = f(c)\, g(c) \int_a^b g(x)\, dx$$

Cancelling g(c) from both sides since it is not 0 gives the result.

□

Now take the taylor remainder $\frac{1}{(n-1)!} \int_0^h (h-u)^{n-1} f^{(n)}(a+u)\, dt$

And assume that $f^{(n)}$ is continuous.

Rewrite the integral as

$$\frac{h^n}{(n-1)!} \int_0^1 (1-t)^{n-1} f^{(n)}(a+th)\, dt$$

By a simple substitution/scaling. Now apply the previous proposition:

The remainder is

$$\frac{h^n}{(n-1)!} f^{(n)}(a+\theta h) \int_0^1 (1-t)^{n-1} dt$$

For some $\theta$ between 0 and 1. But this is just

$$\frac{h^n}{n!} f^{(n)}(a+\theta h)$$

Which is another form of taylor's theorem.

But if we take g=1 and f the rest of the integral we get the remainder is

$$\frac{h^n}{(n-1)!} (1-\theta)^{n-1} f^{(n)}(a+\theta h)$$

For some $\theta$ between 0 and 1.

Of course we can do improper integrals by taking a limit or taking multiple limits. We know this from level 5. As an example,

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \int_{-\infty}^{-1} e^{-x^2} dx + \int_1^{\infty} e^{-x^2} dx + \int_{-1}^{1} e^{-x^2} dx$$

The first 2 terms are finite by comparing to $e^{-|x|}$ which we know has a convergent integral.

Note that just like in the example above, we can test for convergence by comparison or by showing that the ratio between one integral with a known convergent integral approaches something finite. For example, $\frac{x}{x^4+1}$ is $< \frac{1}{x^3}$ so its integral from 1 to infinity converges. In fact it is $\frac{\pi}{8}$ if you want to calculate it.

**Example.** By considering when the limit exists, $\int_0^1 \frac{1}{x^p} dx$ if and only if $p < 1$.

**Example.** If we want to know when $\int_0^{\frac{1}{2}} \frac{1}{x \log^{\mathrm{p}}(x)} dx$, then the substitution $u = \log(x)$ and reducing to the previous example means it converges if and only if p>1.

# 5 Power series

## 5.1 Basic properties (Review)

We say a sequence of functions converges pointwise if for every x, $f_n(x) \to f(x)$ as a sequence of real or complex numbers. Series of functions are just the sums of functions.

For the limit of a sequence of continuous functions to be continuous, we essentially need

$$\lim_{y\to x} \lim_{n\to\infty} f_n(y) = \lim_{y\to x} f(y) = f(x) = \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \lim_{y\to x} f(y)$$

As we know from level 6.4, if we have uniform convergence we can swap these limits (uniform limits of continuous functions are continuous), but swapping limits is not always trivial. It is false in general: For example the limit of

$$f(x) = \begin{cases} nx & x < \frac{1}{n} \\ 1 & \text{Otherwise} \end{cases}$$

on $[0,1]$ is 0 at 0 and 1 everywhere else, which is not continuous but it is a limit of continuous functions.

In fact, it is not even true that

$\lim_{m\to\infty} \lim_{n\to\infty} \frac{m}{m+n} = 0$ but $\lim_{n\to\infty} \lim_{m\to\infty} \frac{m}{m+n} = 1$

We will focus in this course on power series. Recall that the radius of convergence of a power series $c_0 + c_1x + c_2x^2 + \dots$ with $c_i \in \mathbb{C}$ is given by $\frac{1}{\limsup |c_n|^{\frac{1}{n}}}$ and that we have absolute convergence whenever we are strictly inside the radius of convergence.

As an example, if $c_n = n^a$ for any real a, then $|c_n|^{\frac{1}{n}} = n^{\frac{a}{n}} = e^{\frac{a}{n}\log(n)}$. This goes to 1 as n goes to infinity, since n grows faster than log n, or by L'hopital's rule. So the radius of convergence of any such power series is 1.

See levels 4 and 6 for power series properties such as differentiability (which implies continuity) and integrability and all that.

*Sketch of proof that power series are differentiable and integrable (See levels 4 and 6.1 for a more complete proof):*

First we rewrite $b^n - a^n - n(b-a)a^{n-1}$ as $(b-a)^2\left(b^{n-2} + 2ab^{n-3} + ... + (n-1)a^{n-2}\right)$ to make the algebra work in future steps, then we show that the derivative and integral would not affect the radius of convergence using the formula for the radius of convergence (meaning that the reverse derivative always exists so they are integrable), then we show that the "algebraic" derivative and the actual derivative go to 0 as the h in the definition of the derivative goes to 0.

## 5.2 Exponential and trigonometric functions

Let $e^x$ be defined by its power series. Recall from level 4 how we can prove the basic properties $(e^x)' = e^x,\ e^0 = 1,\ e^{a+b} = e^a e^b$.

We will give an alternative easier proof of $e^{a+b} = e^a e^b$.

Let $f(z) = e^{a+b-z}e^z$, then $f'(z) = -e^{a+b-z}e^z + e^{a+b-z}e^z$ by the product and chain rules. Therefore f is constant. Therefore, $e^{a+b-b}e^b = e^{a+b-0}e^0$, so the result follows.

Lets formalize results we already know.

$e^x$ is always positive for real numbers since it cannot cross 0 as if $e^x = 0$ then $e^{-x}$ is undefined. But by the intermediate value theorem, if $e^x$ were ever negative, it would have to cross 0. And we know that $e^x$ is strictly increasing – this is immediate from the series definition, and the derivative is always positive.

Note that $e^x$ for real numbers x indeed hits every real number but 0 because $\lim_{x\to-\infty} e^x = \frac{1}{\lim_{x\to\infty} e^x}$, but we know that $e^x \geq 1 + x$ so it gets arbitrarily large, so the limit tends to 0.

Now that we know that the exp is strictly increasing, we know it has an inverse, which is the logarithm. We knew this before but now we know know it.

By our inverse functions theorems we rederive that the derivative of log is $\frac{1}{x}$.

We want to use taylor series to show that $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln(2)$. However, this is on the edge of the radius of convergence, so we need another lemma. Once we have this, we will know immediately from the arctan series that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}$.

**Theorem.** Suppose we have a power series with radius of convergence R. If $\sum c_n x^n$ converges and $|x| = R$, then the limit as a goes to R along the line arg(a)=arg(x) in the direction away from the origin of $\sum c_n a^n$ is $\sum c_n x^n$

**Why do we want this:** Because we know that taylor series work strictly inside the radius of convergence, but series like $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots = \ln(2)$ and $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} - \cdots = \frac{\pi}{4}$ rely on knowing that the taylor series for $\ln(1+x)$ and $\arctan(x)$ respectively work at the boundary of their radius of convergence. If we can show that the power series that we get from using the log taylor series evaluated at 0.9, 0.99, 0.999, etc that we know are equal to $\ln(1.9), \ln(1.99), \ln(1.999)$ etc converge to $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \ldots$, which clearly converges itself, then it will mean that $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \ldots$ coincides with ln(2) (since ln is continuous there), so the taylor series will work at the edge of the radius of convergence whenever it actually converges there and the function we are expressing as a taylor series is continuous there, and we know already that it works inside the radius of convergence. So that is why we want this theorem.

*Proof.* We will suppose that $R = 1$ and that the point we are considering where the power series converges is the point $x = 1$. This is because we can scale and rotate it as necessary afterwards, but this will simplify calculations.

Therefore we are supposing that $\sum_{n=0}^{\infty} c_n$ converges to a value we will call s and that the power series $\sum c_n x^n$ has radius of convergence 1.

Note that $c_n = \sum_{r=0}^{n} c_r - \sum_{r=0}^{n-1} c_r$. This seems like a complicated way of doing things but it will work out.

Now let x be a real number strictly between 0 and 1.

Now $\sum_{n=0}^{N} c_n x^n = c_0 + \sum_{n=1}^{N} c_n x^n = c_0 + \sum_{n=1}^{N} (\sum_{r=0}^{n} c_r - \sum_{r=0}^{n-1} c_r) x^n$. Now we can do a little trick: We hav the following terms in our sum:

| | |
|---|---|
| $\sum_{r=0}^{1} c_r x$ | $-\sum_{r=0}^{0} c_r x$ |
| $\sum_{r=0}^{2} c_r x^2$ | $-\sum_{r=0}^{1} c_r x^2$ |
| $\sum_{r=0}^{3} c_r x^3$ | $-\sum_{r=0}^{2} c_r x^3$ |
| ... | ... |
| $\sum_{r=0}^{N} c_r x^N$ | $-\sum_{r=0}^{N-1} c_r x^{N+1}$ |

Now we will separate out the top-right and bottom-left terms, and simplify the top right term to $c_0 x$. Then the sum of the rest of the terms (add each term to the one directly to the bottom-right of it) is exactly $\sum_{n=1}^{N-1} \sum_{r=0}^{n} c_r \left(x^n - x^{n+1}\right)$. So

$$\sum_{n=0}^{N} c_n x^n = c_0 - c_0 x + \sum_{r=0}^{N} c_r x^N + \sum_{n=1}^{N-1} \sum_{r=0}^{n} c_r \left(x^n - x^{n+1}\right)$$

Therefore

$$\sum_{n=0}^{N} c_n x^n = c_0(1-x) + \sum_{r=0}^{N} c_r x^N + \sum_{n=1}^{N-1} \sum_{r=0}^{n} c_r (1-x) x^n$$

But then we can put the term $c_0(1-x)$ into the sum on the right, ie we get

$$\sum_{n=0}^{N} c_n x^n = \sum_{r=0}^{N} c_r x^N + (1-x) \sum_{n=0}^{N-1} \sum_{r=0}^{n} c_r x^n$$

. By the hypothesis of the theorem, the right hand side converges. Since $\sum_{r=0}^{N} c_r$ converges, it is bounded. Since $x^N$ goes to 0 as N gets large, $\sum_{r=0}^{N} c_r x^N$ goes to 0 since $\sum_{r=0}^{N} c_r$ is bounded. Therefore when we take the limit as N goes to infinity that term vanishes and we get

$$\sum_{n=0}^{\infty} c_n x^n = (1-x) \sum_{n=0}^{\infty} \sum_{r=0}^{n} c_r x^n$$

The sum on the right converges since differs from a convergent sum by something that approaches 0. Now our goal is to show that as x goes to 1 from below, $\sum_{n=0}^{\infty} c_n x^n$ approaches s. Note that since x<1, we can safely say that $(1-x) \sum_{n=0}^{\infty} x^n = 1$ (geometric series or generalized binomial theorem).

Therefore we can subtract s from both sides of the equation we deduced above to get that

$$\sum_{n=0}^{\infty} c_n x^n - s = (1-x) \sum_{n=0}^{\infty} \left( \sum_{r=0}^{n} c_r - s \right) x^n$$

Therefore if we can show that the right hand side tends to 0, we will know that the left hand side tends to 0 so we will be done. Now lets pick a number $\varepsilon$ as small as we like then pick an M such that for all $n \geq M$, $|\sum_{r=0}^{n} c_r - s| < \frac{\varepsilon}{2}$, possible by definition of summing to infinity.

Then we will write $\sum_{n=0}^{\infty} \left(\sum_{r=0}^{n} c_r - s\right) x^n$ as $\sum_{n=0}^{M-1} \left(\sum_{r=0}^{n} c_r - s\right) x^n + \sum_{n=M}^{\infty} \left(\sum_{r=0}^{n} c_r - s\right) x^n$

$$\sum_{n=0}^{\infty} c_n x^n - s = (1-x) \sum_{n=0}^{M-1} \left(\sum_{r=0}^{n} c_r - s\right) x^n + \sum_{n=M}^{\infty} \left(\sum_{r=0}^{n} c_r - s\right) x^n$$

So by the triangle inequality

$$\left|\sum_{n=0}^{\infty} c_n x^n - s\right| \leq |1-x| \sum_{n=0}^{M-1} \left|\sum_{r=0}^{n} c_r - s\right| |x|^n + \sum_{n=M}^{\infty} \left|\sum_{r=0}^{n} c_r - s\right| |x|^n$$

$$\left|\sum_{n=0}^{\infty} c_n x^n - s\right| \leq |1-x| \sum_{n=0}^{M-1} \left|\sum_{r=0}^{n} c_r - s\right| |x|^n + \sum_{n=M}^{\infty} \frac{\varepsilon}{2} |x|^n$$

$$\left|\sum_{n=0}^{\infty} c_n x^n - s\right| \leq |1-x| \sum_{n=0}^{M-1} \left|\sum_{r=0}^{n} c_r - s\right| |x|^n + |1-x| \frac{\varepsilon}{2} \frac{|x|^M}{1-|x|}$$

(last line by geometric series). Since $0 < x < 1$, $|x|^M < 1$, so the last term is ast most $\frac{\varepsilon}{2}$. Since M does not depend on x and x is a positive real number less than 1, the term $|1-x| \sum_{n=0}^{M-1} |\sum_{r=0}^{n} c_r - s| \, |x|^n$ is at most $|1-x| \sum_{n=0}^{M-1} |\sum_{r=0}^{n} c_r - s|$, which is just $|1-x| \, C$ for some constant C. Therefore this tends to 0 as x gets close enough to 1, in particular it is eventually less than $\frac{\varepsilon}{2}$. Therefore, for x close enough to 1,

$$\left|\sum_{n=0}^{\infty} c_n x^n - s\right| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

This means the power series can be made arbitrarily close to s by making x sufficiently close to 1 from the left of 1 on the number line. This completes the proof of the theorem.

$\square$

Now using our stuff about exponentials and logarithms we can define $x^a = e^{a\log(x)}$ and derive the usual properties of exponents.

Recall that by repeated l'hopital's rule we can formalize the "exponentials beat powers" and "powers beat logarithms" ideas.

Recall that $\sin(x) = \frac{e^{ix} - e^{-ix}}{2}$ and $\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$, and that all of the trig identities we know follow from the properties of the exponential and taking real and imaginary parts.

We want to see another way of seeing that these are periodic, i.e. $\sin(x) = \sin(x + 2\pi)$ and similarly for cos.

To do this, we will prove that there is some smallest $w$ (This $w$ is the number commonly referred to as $\pi$) such that $\cos\left(\frac{w}{2}\right) = 0$.

We will prove that $\sin\left(\frac{w}{2}\right) = 1$, and that sin and cos are periodic with period 2w. We will prove also that adding w to sin or cos makes the result minus. We will prove also that $\sin\left(x + \frac{w}{2}\right) = \cos(x)$ and that $\cos\left(x + \frac{w}{2}\right) = -\sin(x)$.

Ok so here is how we do this:

We can calculate by numerical calculation and an easy comparison to series that $\cos(2) < 0$ since the first 4 terms of the taylor series give $-\frac{19}{45}$ and summing the absolute value of the remaining terms, ie $\sum_{k=4}^{\infty} \frac{2^{2k}}{(2k)!}$ is bounded above by $\sum_{k=4}^{\infty} \frac{2^{2k}}{8^k}$ which is $\frac{1}{8}$ (since $8^4 < (2*4)!$ And it is true for k>4 by induction) and we know trivially $\cos(0) = 1$. So by the intermediate value theorem, we know that there is some first place cos attains 0 between 0 and 2. We will call this $\frac{w}{2}$.

Now we get $\sin\left(\frac{w}{2}\right) = 1$ since sin was increasing on $\left(0, \frac{w}{2}\right)$ by the derivative. By the addition formula, get get $\sin\left(x + \frac{w}{2}\right) = \cos(x)$ and that $\cos\left(x + \frac{w}{2}\right) = -\sin(x)$. So periodicity follows from repeatedly applying these. We get periodicity of the exponential as well.

**Proposition.** 2w is the perimeter of the unit circle.

*Proof.* We want to consider the length of the curve $e^{it}$ as t goes from 0 to 2w. By the arc length formula, we will have $\int_0^{2w} \left|\frac{d}{dt}e^{it}\right| dt = \int_0^{2w} 1dt = 2w$. This is the length because we are **defining** length this way.

□

# 6 Uniform convergence

## 6.1 Basic properties

Recall that we defined uniform convergence in level 6. Roughly speaking, it says that a function not only converges everywhere, but that there is a minimum rate of convergence across the domain. Here are the quantifiers:

Functions $f_n$ defined on $E$ going to $\mathbb{R}$ (or $\mathbb{C}$) converge to $f$ uniformly if

$$(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall n > N) \sup_{x \in E} |f_n(x) - f(x)| < \varepsilon$$

**Example.** Let $f_n : \mathbb{R} \to \mathbb{R}$ be a sequence of functions defined by $f_n(x) = \frac{x}{n}$. Then $f_n \to 0$ pointwise. However, the convergence is not uniform because $|f_n(x) - 0| = \frac{|x|}{n}$ which can be made arbitrarily large for each fixed $n$. However, if we restrict $f_n$ to a bounded domain, say $[-a, a]$ then the convergence becomes uniform because

$$\sup |f_n(x) - 0| = \frac{|x|}{n} \leq \frac{a}{n}$$

So given $\varepsilon$ pick $N$ with $N > \frac{a}{\varepsilon}$ then we're done.

**Definition.** Let E be a set. A sequence $f_n : E \to \mathbb{R}$ of functions is uniformly Cauchy if

$$(\forall \varepsilon > 0)(\exists N)(\forall m, n > N) \sup_{x \in E} |f_n(x) - f_m(x)| < \varepsilon$$

**Exercise.** Verify that this definition coincides with what you would expect "uniformly Cauchy" to mean.

**Theorem.** Let $f_n : E \to \mathbb{R}$ be a sequence of functions. Then $f_n$ converges uniformly if and only if $f_n$ is uniformly Cauchy.

*Proof.* First suppose that $f_n \to f$ uniformly. Given $\varepsilon$ there exists $N$ such that

$$(\forall n > N) \sup_{x \in E} |f_n(x) - f(x)| < \frac{\varepsilon}{2}$$

Then if $n, m > N$ and $x \in E$ we have, by the triangle inequality,

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f_m(x) - f(x)| < \varepsilon$$

Since this is true for all $x$, taking the supremum on the left hand side gives the desired result.

Now suppose $f_n$ is uniformly Cauchy, then $f_n(x)$ is a Cauchy sequence for all $x$ so it converges. Let $f(x)$ be the pointwise limit, then we want to show that the convergence is uniform. Fix $\varepsilon > 0$ and choose $N$ such that whenever $n, m > N$ and $x \in E$ we have $|f_n(x) - f_m(x)| < \varepsilon$. Letting $m \to \infty$ we have $|f_n(x) - f(x)| \leq \varepsilon$, then we are done since this is independent of $x$.

□

**Theorem.** Let $f_n, f : [a, b] \to \mathbb{R}$ be Riemann integranle and suppose $f_n \to f$ uniformly. Then $\int_a^b f_n(t)\,dt \to \int_a^b f(t)\,dt$. (We will prove in the next theorem that this actually implies $f$ is integrable)

*Idea.* Uniform convergence forces the functions in the sequence to eventually be in arbitrarily small bands around the function, and if the band has width $\varepsilon$, then the integral difference between $f_n$ and $f$ is bounded above in absolute value by $\varepsilon(b - a)$.

*Proof.*

$$\left|\int_a^b f_n - \int_a^b f\right| = \left|\int_a^b f_n - f\right| \leq \int_a^b |f_n - f| \leq (b - a)\sup|f_n - f| \to 0$$

□

**Theorem.** Let $f_n : [a, b] \to \mathbb{R}$ be uniformly bounded and integrable for all $n$. Then if $f_n \to f$ uniformly then $f$ is bounded and integrable.

*Proof.* Let $c_n = \sup_{[a,b]}|f_n - f|$. Then uniform convergence implies that $c_n \to 0$ and for each $x$ we have

$$f_n(x) - c_n \leq f(x) \leq f_n(x) + c_n$$

Since $f_n$ is bounded this implies that $f$ is bounded by $\sup|f_n| + c_n$. Also, for any $x, y \in [a, b]$ we know

$$f(x) - f(y) \leq (f_n(x) - f_n(y)) + 2c_n$$

Therefore for any partition $P$ we have

$$U(P, f) - L(P, f) \leq U(P, f_n) - L(P, f_n) + 2(b - a)c_n$$

Given $\varepsilon > 0$ first choose $n$ such that $2(b - a)c_n < \frac{\varepsilon}{2}$ (possible as $c_n \to 0$) then choose $P$ such that $U(P, f_n) - L(P, f_n) < \frac{\varepsilon}{2}$ (possible as $f_n$ is integrable), then for this partition $U(P, f) - L(P, f) < \varepsilon$.

□

It is an easy check that if $f_n \to f$ uniformly and $g_n \to g$ uniformly then for any $a, b \in \mathbb{R}$ we have $af_n + bg_n \to af + bg$ uniformly.

**Proposition.** Suppose $f_n \to f$ uniformly on $E$ and $g : E \to \mathbb{R}$ is bounded, then $gf_n \to gf$ uniformly.

*Proof.*

$$\sup_E|g(x)f_n(x) - g(x)f(x)| \leq \sup_E|g(x)|\sup_E|f_n(x) - f(x)|$$

The right hand term is a constant multiplied by a sequence that approaches 0.

□

## 6.2 Uniform convergence and differentiation

**Example.** In the example in level 4 of $\frac{\cos(nx)}{n}$ where you could not swap a derivative with a limit, note that in fact the convergence was uniform. So a derivative of a limit need not equal a limit of a derivative even if the convergence is uniform.

**Example.** If $f_n(x) = \sqrt{\frac{1}{n} + x^2}$ then it is easy to see that each $f_n$ is differentiable and that they converge uniformly to $|x|$ which is not differentiable, so the uniform limit of differentiable functions might not even be differentiable.

However, there is a theorem that makes things play nice.

**Theorem.** Let $f_n : [a, b] \to \mathbb{R}$ be a sequence of functions differentiable on $[a, b]$ where at the end points the one-sided derivatives exist. Suppose the following hold:

1. For some $c \in [a, b]$, $f_n(c)$ converges.

2. The sequence of derivatives $f'_n$ converges uniformly on $[a, b]$.

Then $f_n$ converges uniformly on $[a, b]$ and its limit is differentiable with derivative $\lim f'_n$

*Idea.* If you don't understand what I'm trying to say here then don't worry and just read the proof. The heuristic for why these conditions are enough and where they even come from is that by essentially the reverse idea as the integration proof, if the derivative goes into a thin band then the function goes into a thin sector by the mean value theorem, and $c$ can be used as an anchor point so that convergence can't be slower than within a very small sector around the $c$ point where we know it is well behaved.

*Proof.* To show that $f_n$ converges uniformly we will prove it is uniformly Cauchy. We seek $N$ such that $(n, m > N) \implies \sup |f_n - f_m| < \varepsilon$. Fix $x \in [a, b]$ and suppose $x \neq c$ (since it is easy to see that if $f_n$ converges uniformly everywhere else adding one point where it converges won't change that). By the mean value theorem on $f_n - f_m$, we have

$$(f_n - f_m)(x) - (f_n - f_m)(c) = (x - c)(f'_n - f'_m)(t)$$

for some $t \in (\min(x, c), \max(x, c))$. Easy check: The above inequality is valid whether $x > c$ or $x < c$.

We then get

$$(f_n - f_m)(x) - (f_n - f_m)(c) = (x - c)(f'_n - f'_m)(t)$$

Therefore

$$(f_n - f_m)(x) \leq (f_n - f_m)(c) + (b - a)(f'_n - f'_m)(t)$$

$$\sup_{x \in [a,b]} |f_n(x) - f_m(x)| \leq |f_n(c) - f_m(c)| + \sup_{t \in [a,b]} |f'_n(t) - f'_m(t)|$$

Note: To see why taking suprema on both sides preserves inequalities like that, consider doing it first to one side then the other.

Note that by the hypothesis of the theorem the 2 terms on the right tend to 0 as $N \to \infty$, which implies that $f_n$ is uniformly Cauchy and thus uniformly convergent. Let $f = \lim f_n$ and let $f'_n \to h$. For each $y \in [a, b]$ define

$$g_n(x) = \begin{cases} \frac{f_n(x) - f_n(y)}{x - y} & x \neq y \\ f'_n(y) & x = y \end{cases}$$

Therefore $f_n$ is differentiable at $y$ if and only if $g_n$ is continuous at $y$. Also define

$$g(x) = \begin{cases} \frac{f(x) - f(y)}{x - y} & x \neq y \\ h(y) & x = y \end{cases}$$

Then $f$ is differentiable with derivative $h$ at $y$ if and only if $g$ is continuous at $y$. This would follow if $g_n \to g$ uniformly as we know the $g_n$ are all continuous. To show that the convergence is uniform we will show that the sequence is uniformly Cauchy. For $x \neq y$ we have

$$g_n(x) - g_m(x) = \frac{(f_n - f_m)(x) - (f_n - f_m)(y)}{x - y} = (f'_n - f'_m)(t)$$

for some $t$ between $x$ and $y$. This also holds if $x = y$ since $g_n(y) - g_m(y) = f'_n(y) - f'_m(y)$ by definition. Let $\varepsilon > 0$. Then since $f'$ converges uniformly there is some $N$ such that for all $x \in [a, b]$ we have

$$|g_n(x) - g_m(x)| \leq \sup |f'_n - f'_m| < \varepsilon$$

since the derivatives converge uniformly. Therefore $g_n$ converges uniformly as we set out to prove.

Since the limit function $g$ is continuous at $y$, the derivative of $f$ at y must be none other than $\lim (f'(y))$

$\square$

### 6.3 Uniform convergence and series

**Definition.** A series of functions $\sum g_n$ is said to converge absolutely uniformly if $\sum |g_n|$ converges uniformly.

**Proposition.** Let $g_n : E \to \mathbb{R}$. Then if $\sum g_n$ converges absolutely uniformly then it converges uniformly.

*Proof.* Let $f_n = \sum_{j=1}^{n} g_j$ and $h_n = \sum_{j=1}^{n} |g_j|$. Then for $n > m$ we have

$$|f_n(x) - f_m(x)| = \left| \sum_{j=m+1}^{n} g_j(x) \right| \leq \sum_{j=m+1}^{n} |g_j(x)| = |h_n(x) - h_m(x)|$$

Therefore we have

$$\sup |f_n(x) - f_m(x)| \leq \sup |h_n(x) - h_m(x)| \to 0 \text{ by hypothesis}$$

So the result follows as the sequence is uniformly Cauchy.

□

Note that absolute convergence and uniform convergence does not imply absolute uniform convergence. A counterexample is the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} x^n$$

This converges absolutely for every $x \in (0,1)$ and it converges uniformly there (Hint: Alternating series test). However, it does not converge absolutely uniformly (Since $\sum \frac{1}{n}$ diverges and by making $x$ sufficiently close to 1 we can make the sum of the absolute values get arbitrarily close to any arbitrarily large partial sum of $\sum \frac{1}{n}$)

**Theorem.** (Weierstrass M-test) Let $g_n : E \to \mathbb{R}$ be a sequence of functions and suppose that there exists a sequence $M_n$ such that for all $n$ we have $\sup_{x \in E} |g_n(x)| \leq M_n$. Then if $\sum M_n$ converges then $\sum g_n$ converges absolutely uniformly.

*Proof.* Let $f_n = \sum_{j=1}^{n} |g_j|$ be the partial sums. Then for $n > m$ we have

$$|f_n(x) - f_m(x)| = \sum_{j=m+1}^{n} |g_j(x)| \leq \sum_{j=m+1}^{n} M_j \xrightarrow[n,m \to \infty]{} 0$$

so $f_n$ is uniformly Cauchy and therefore uniformly convergent.

□

# 7 Useful theorems

### 7.1 Lebesgue's criterion for Riemann interability

**Theorem.** (Lebesgue's Criterion) Recall that a function is Riemann integrable if overestimating or underestimating it with rectangles lets us get as close as we want and we proved that in 1D this is satisfied for continuous functions. Let Q be a closed bounded rectangle in $\mathbb{R}^n$ and let $f : Q \to \mathbb{R}$ be a bounded function. Let D be the set of points of Q such that f is not continuous. Then the converse to the theorem we proved in level 6.2 holds: $f$ is Riemann integrable over Q if D has measure 0.

*Proof.* Use the fact (level 6.4) that closed and bounded sets have the property that every open cover has a finite subcover - more on this later.

Pick M such that $|f(x)| < M$ for x in Q (possible by boundedness).

First, assume D has measure 0. Fix $\varepsilon > 0$ and define $\varepsilon' := \frac{\varepsilon}{2M + 2\text{vol}(Q)}$. Cover D by countably many open rectangles $Q_1, Q_2, Q_3, \ldots$ with total volume less than $\varepsilon'$, possible as D has measure 0. For each point $a$ in $Q \setminus D$ pick an open rectangle $Q_a$ containing a such that (by continuity of f at $a$) $|f(x) - f(a)| < \varepsilon'$ for $x \in Q_a \cap Q$. Since $Q$ is closed and bounded we can pick a finite subcover of all the rectangles from points in $Q \setminus D$ and $D$, which will be given by $Q_1, Q_2, Q_3, \dots Q_k, Q_{a_1}, Q_{a_2}, \dots, Q_{a_n}$, and these rectangles cover $Q$. The first k need not cover D but it turns out that that will not matter. Set $Q'_j := Q_{a_j}$, then we have a cover of k+n rectangles of Q by $Q_{1\dots k}, Q'_{1\dots n}$, where $Q_{1\dots k}$ has total volume less than $\varepsilon'$. The rectangles $Q'_j$ satisfy for $x, y$ in $Q'_j \cap Q$

$$|f(x) - f(y)| \leq |f(x) - f(a_j)| + |f(a_j) - f(y)| \leq 2\varepsilon'$$

For convenience replace each rectangle by its intersection with Q. Now using the end points of each rectangle define a partition P of Q into a grid such that we cut it at each coordinate, i.e. the smallest grid partition such that each rectangle is made up of grid lines. Divide the rectangles in the partition into R so that each rectangle in R lies in one of the Q rectangles and R' such that each rectangle in R' lies in one of the Q' rectangles, or alternatively R' is everything not in R. Then we have

$$\sum_{r\in R} \left(\max_r (f) - \min_r (f)\right) \mathrm{vol}(r) \leq 2M \sum_{r\in R} \mathrm{vol}(r)$$

$$\sum_{r\in R'} \left(\max_r (f) - \min_r (f)\right) \mathrm{vol}(r) \leq 2\varepsilon' \sum_{r\in R} \mathrm{vol}(r)$$

Both by definition.

Now $\sum_{r\in R} \mathrm{vol}(r) < \varepsilon'$ because the total volume of the Q rectangles is less than that by assumption, and $\sum_{r\in R} \mathrm{vol}(r) \leq \mathrm{vol}(Q)$. Thus

$$\sum_{r\in P} \left(\max_r (f) - \min_r (f)\right) \mathrm{vol}(r) = 2M \sum_{r\in R} \mathrm{vol}(r) + 2\varepsilon' \sum_{r\in R'} \mathrm{vol}(r) \leq 2M\varepsilon' + 2\mathrm{vol}(Q)\,\varepsilon' = \varepsilon$$

□

**Corollary.** If $f : [a, b] \to [A, B]$ is integrable and $g : [A, B] \to \mathbb{R}$ is continuous then $g \circ f : [a, b] \to \mathbb{R}$ is integrable

**Definition.** A vector-valued function $f : [a, b] \to \mathbb{R}^n$ is integrable if each of its components are integrable, and the integral of $f$ is defined so that each component is the value of the integral of that component.

**Corollary.** Let $f : [a, b] \to \mathbb{R}^n$ be integrable, then the function $\|f\| : [a, b] \to \mathbb{R}$ defined by $\|f\|(x) = \sqrt{\sum_{j=1}^{n} f_j^2(x)}$ is integrable and $\left\|\int_a^b f(x)\,dx\right\| \leq \int_a^b \|f\|(x)\,dx$

*Proof.* Integrability follows from the theorem above. To show the inequality set $v_j = \int_a^b f_j(x)\,dx$ so that

$$\|v\|^2 = \sum_{j=1}^{n} v_j^2 = \sum_{j=1}^{n} v_j \int_a^b f_j(x)\,dx = \int_a^b v.f(x)\,dx$$

Then Cauchy-Schwarz implies

$$\leq \int_a^b \|v\| \, \|f\|(x)\,dx = \|v\| \int_a^b \|f\|(x)\,dx$$

Dividing both sides by $\|v\|$ (Note: if $\|v\|$ is 0 then the result is trivially true) gives the result.

□

## 7.2 Weierstrass approximation theorem

Recall that uniform continuity is like continuity but for each $\varepsilon$ there is a $\delta$ that works for all $x$ simultaneously and not just one for each $x$ that may depend on $x$. Recall also that a continuous function on a closed bounded subset of $\mathbb{R}^n$ is uniformly continuous.

**Theorem.** (Weierstrass approximation theorem) Let $f : [0, 1] \to \mathbb{R}$ be continuous. Then there exists a sequence of polynomials $p_n$ such that $p_n \to f$ uniformly.

*Proof.* We claim that $p_n$ can be given explicitly by the formula

$$p_n(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \begin{pmatrix} n \\ k \end{pmatrix} x^k (1-x)^{n-k}$$

Note this is not unique, since we could (for example) add $\frac{1}{n}$ to each and if the above sequence works that would still converge uniformly.

For the proof define

$$p_{n,k}(x) = \begin{pmatrix} n \\ k \end{pmatrix} x^k (1-x)^{n-k}$$

Note that for all $x \in [0,1]$, $p_{n,k}(x) \geq 0$. By the binomial theorem we have that $\sum_{k=0}^{n} p_{n,k}(x) = 1$. Differentiating the binomial theorem with respect to $x$ gives, for each $y$ fixed,

$$\sum_{k=0}^{n} \begin{pmatrix} n \\ k \end{pmatrix} kx^{k-1} y^{n-k} = \frac{d}{dx}(x+y)^n = n(x+y)^{n-1}$$

Putting $y = 1-x$ into the above and multiplying it by $x$ gives

$$\sum_{k=0}^{n} \begin{pmatrix} n \\ k \end{pmatrix} kx^k y^{n-k} = nx = \sum_{k=0}^{n} kp_{n,k}$$

Differentiating again gives

$$\sum_{k=0}^{n} k(k-1) p_{n,k} = n(n-1)x^2$$

Adding the 2 results and rearranging gives

$$\sum_{k=0}^{n} k^2 p_{n,k} = n^2x^2 + nx(1-x)$$

Using these results and expanding gives

$$\sum_{k=0}^{n} (nx-k)^2 p_{n,k}(x) = n^2x^2 - nx(2nx) + n^2x^2 + nx(1-x) = nx(1-x)$$

Now pick $\varepsilon > 0$. Since $f$ is continuous on $[0,1]$ it is uniformly continuous so there is a $\delta$ such that

$$|x-y| < \delta \implies |f(x) - f(y)| < \frac{\varepsilon}{2}$$

Note that since $\sum_{k=0}^{n} p_{n,k}(x) = 1$, we have $f(x) = \sum p_{n,k}(x) f(x)$ for each fixed $x$. So write

$$|p_n(x) - f(x)| = \left| \sum_{k=0}^{n} \left( f\left(\frac{k}{n}\right) - f(x) \right) p_{n,k}(x) \right| \leq \sum_{k=0}^{n} \left| f\left(\frac{k}{n}\right) - f(x) \right| p_{n,k}(x)$$

since $p_{n,k}$ is non-negative

$$= \sum_{k:\left|x-\frac{k}{n}\right|<\delta} \left( \left| f\left(\frac{k}{n}\right) - f(x) \right| p_{n,k}(x) \right) + \sum_{k:\left|x-\frac{k}{n}\right|\geq\delta} \left( \left| f\left(\frac{k}{n}\right) - f(x) \right| p_{n,k}(x) \right)$$

where $k$ is quantified over $\{0,1,2,...,n\}$ in the line above.

$$\leq \frac{\varepsilon}{2} \sum_{k=0}^{n} p_{n,k}(x) + 2\sup_{[0,1]} |f| \sum_{k:\left|x-\frac{k}{n}\right|\geq\delta} p_{n,k}(x)$$

$$\leq \frac{\varepsilon}{2} + 2\sup_{[0,1]} |f| \frac{1}{\delta^2} \sum_{k:\left|x-\frac{k}{n}\right|\geq\delta} \left( x - \frac{k}{n} \right)^2 p_{n,k}(x)$$

This last line is because of the $\left|x - \frac{k}{n}\right| \geq \delta$ thing.

$$\leq \frac{\varepsilon}{2} + 2 \sup_{[0,1]} |f| \frac{1}{\delta^2} \sum_{k=0}^{n} \left(x - \frac{k}{n}\right)^2 p_{n,k}(x) = \frac{\varepsilon}{2} + 2 \sup_{[0,1]} |f| \frac{1}{n^2 \delta^2} \sum_{k=0}^{n} (nx - k)^2 p_{n,k}(x)$$

$$= \frac{\varepsilon}{2} + 2 \sup_{[0,1]} |f| \frac{1}{n^2 \delta^2} nx(1-x) \leq \frac{\varepsilon}{2} + \frac{\sup_{[0,1]} |f|}{n\delta^2}$$

Thus we can pick $n$ large enough such that this is $< \varepsilon$.

$\square$

# 8 Normed spaces

## 8.1 Basic properties of normed spaces

**Definition.** (Normed space) Let $V$ be a real or complex vector space. A norm on $V$ is a function $\|\cdot\| : V \to \mathbb{R}_{\geq 0}$ satisfying

1. $\|x\| = 0$ if and only if $x = 0$

2. $\|\lambda x\| = |\lambda| \|x\|$ for all $\lambda$

3. $\|x + y\| \leq \|x\| + \|y\|$

Think of this as a way of assigning length to a vector.

**Example.** On $\mathbb{R}^n$ we can consider the usual norm $\|x\|_2 = \sqrt{\sum x_i^2}$. To see that this satisfies the triangle inequality note that by Cauchy-Schwarz we have

$$\|x + y\|_2^2 = \|x\|_2^2 + \|y\|_2^2 + 2(x.y) \leq \|x\|_2^2 + \|y\|_2^2 + 2\|x\|_2 \|y\|_2 = (\|x\|_2 + \|y\|_2)^2$$

In general, for $1 \leq p < \infty$ we can define a norm as

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}}$$

However, unless $p = 1$ or $p = 2$ it is difficult to check the triangle inequality, therefore in this course we will note that those are the only ones we know are actually norms. We can define the infinity norm by

$$\|x\|_\infty = \max(|x_i|)$$

Roughly speaking the infinity convention is because as $p$ gets larger we note that $\|x\|_\infty \leq \|x\|_p \leq \|x\|_\infty n^{\frac{1}{p}}$ which implies that $\|x\|_p \to \|x\|_\infty$.

There are more examples, in fact there are infinite dimensional examples.

**Example.** Consider the set of sequences $(x_k)$ such that $\sum |x_k|^p < \infty$. Then we call this set $l_p$ with norm $\sum |x_k|^p$. However, we only know that the triangle inequality holds in $l_1$ and $l_2$ by passing the limit from the finite case.

**Example.** The set of bounded sequences $(x_k)$ with norm $\sup |x_k|$ is a normed space. This is called $l_\infty$

**Example.** Define $C([a, b])$ as the set of functions which are continuous on $[a, b]$, then there are a few norms we can give it.

We can define the $L^1$ norm by

$$\|f\|_1 = \int_a^b |f| \, dx$$

We can define the $L^2$ norm by

$$\|f\|_2 = \left(\int_a^b |f|^2 \, dx\right)^{\frac{1}{2}}$$

In later courses we will see that 2 can be replaced with $p$ for $1 < p < \infty$. In the case of the $L^2$ norm it satisfies the triangle inequality as we will prove that shortly. As for the norm being 0 implying $|f| = 0$, we'll do that after.

We also have the $L^\infty$ norm, or supremum norm, or uniform norm defined by $\|f\|_\infty = \sup |f|$.

**Lemma.** Let $f, g \in C[a, b]$ be non-negative, then

$$\int_a^b fg dx \leq \sqrt{\int_a^b f^2 dx \int_a^b g^2 dx}$$

*Proof.* Follows from Cauchy-Schwarz on real vector spaces (see Vectors and Matrices) with the inner product on $C[a, b]$ given by $\langle f, g\rangle = \int_a^b fg dx$

□

**Proposition.** The $L^2$ norm satisfies the triangle inequality.

*Proof.* We use the previous lemma. Note that

$$\|f+g\|_2 = \sqrt{\int_a^b |f+g|^2 \, dx} \leq \sqrt{\int_a^b |f|^2 + |g|^2 + 2|f||g| \, dx}$$

$$\leq \sqrt{\int_a^b |f|^2 + |g|^2 \, dx + 2\sqrt{\int_a^b |f|^2 \, dx \int_a^b |g|^2 \, dx}}$$

$$= \sqrt{\int_a^b |f|^2 \, dx} + \sqrt{\int_a^b |g|^2 \, dx} = \|f\|_2 + \|g\|_2$$

□

Note that we might want to define $L^1$ to be the set of functions such that $\int |f|$ exists, but this is difficult since $f$ could be non-zero but the norm could be 0. We do this properly in the Probability and Measure course. I ought to prove we don't have this issue for continuous functions, so I'll do that now.

**Proposition.** Let $f$ be continuous and non-negative and non-zero on $[a, b]$, then $\int_a^b f$ is non-zero

*Proof.* Note that there exists an interior point, $x \in (a, b)$ such that $f(x) > 0$, say $f(x) = \varepsilon > 0$. Then there exists a $\delta > 0$ by continuity such that $|y - x| < \delta \implies f(y) > \frac{\varepsilon}{2}$ and the contribution to the integral from the set of $y$ with $|y - x| < \delta$ is $\varepsilon\delta > 0$.

□

**Definition.** Let $V$ be a normed space. Two norms $\|\cdot\|$ and $\|\cdot\|'$ are said to be Lipschitz equivalent if there are real constants $0 < a < b$ such that for all $x$ we have

$$a\|x\| \leq \|x\|' \leq b\|x\|$$

Easy check: This is an equivalence relation on the set of all norms on $V$.

## 8.2 Open and closed sets

**Definition.** (Open ball) Let $(V, \|\cdot\|)$ be a normed space, and let $a \in V$ and let $r > 0$, then define the open ball centered at $a$ with radius $r$ as

$$B_r(a) = \{x \in V : \|x - a\| < r\}$$

A closed ball is similarly defined as

$$\overline{B_r(a)} = \{x \in V : \|x - a\| \leq r\}$$

Now note that Lipschitz equivalence becomes equivalent to existence of $0 < a < b$ such that

$$B_{\frac{1}{b}}(0) \subseteq B_1'(0) \subseteq B_{\frac{1}{a}}(0)$$

Where $B'$ is the ball with respect to $\|\cdot\|'$ and $B$ is the ball with respect to $\|\cdot\|$.

**Example.** In $\mathbb{R}^2$ the norms $\|\cdot\|_2$ and $\|\cdot\|_\infty$ are equivalent by Figure 4:

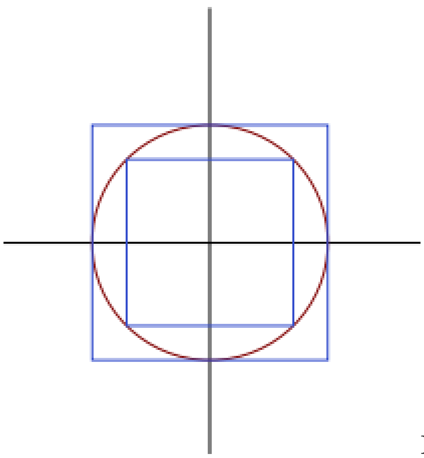

Figure 4

Where the inner blue balls are for $\|\cdot\|_\infty$ and the circle is from $\|\cdot\|_2$.

Similar logic shows that this works in $\mathbb{R}^n$.

**Example.** Let $V = C([0,1])$. Then clearly we have the bound $\|f\|_1 \leq \|f\|_\infty$ (This is just because $\int_0^1 |f| \leq \sup |f|$). However, there is no constant $M$ such that $\|f\|_\infty \leq M \|f\|_1$. To see why, consider the spike functions

$$f_n(x) = \max\left(0, 1 - n\left|x - \frac{1}{n}\right|\right)$$

These are a spike of width $\frac{2}{n}$ starting at the origin.

Similarly by considering the sequence $x^n = (1, 1, 1, ...1, 0, 0, ...)$ with n 1's we see that we cannot say $\|(x_k)\|_{l_2} \leq M \|x_k\|_\infty$ for any $M$, even though it is clear that $\|x_k\|_\infty \leq \|(x_k)\|_{l_2}$ for a similar reason to above.

**Definition.** A set in a normed space is bounded if there is $R > 0$ such that the set is contained in $B_R(0)$

**Definition.** A sequence in a normed space is said to converge if

$$(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall k \geq N)\|x_k - x\| < \varepsilon$$

Note that if two norms are equivalent they agree on what is bounded and what converges.

**Proposition.** In a normed space we have

1. $x_k \to x$ and $x_k \to y$ implies $x = y$

2. $x_k \to x$ implies $ax_k \to ax$ for every $a$.

3. If $x_k \to x$ and $y_k \to y$ then $x_k + y_k \to x + y$

*Proof.* 1. For all $\varepsilon > 0$ there is a k such that

$$\|x - y\| \leq \|x - x_k\| + \|y - x_k\| < 2\varepsilon$$

The only way this can be true for all $\varepsilon$ is if $\|x - y\| = 0$.

2. $\|ax_k - ax\| = |a| \, \|x_k - x\| \to 0$

3. $\|(x_k + y_k) - (x + y)\| \leq \|x_k - x\| + \|y_k - y\| \to 0$

$\square$

Note that in $\mathbb{R}^n$ it is easy to see that the $\|\cdot\|_1$ and $\|\cdot\|_\infty$ norms are equivalent norms with constants $\frac{1}{n}$ and 1: The reason is that-

If all absolute values of the coordinates are $\leq \frac{1}{n}$ their sum is $\leq 1$

If the sum is $\leq 1$ they are all $\leq 1$.

Therefore the $\|\cdot\|_1$ and $\|\cdot\|_\infty$ norms are equivalent and therefore equivalent to the $\|\cdot\|_2$ norm (We will show soon that any norms on a finite dimensional vector space are equivalent).

From this, it follows that:

If the coordinates all converge to 0 then their maximum converges to 0 so we have convergence in the $\|\cdot\|_\infty$ norm. The converse is also true, so convergence in $\|\cdot\|_\infty$ and $\|\cdot\|_2$ and $\|\cdot\|_1$ is equivalent to coordinate-wise convergence.

**Example.** Note that in $l_\infty$ the set of sequences $x^n = (0, 0, ..., 0, 1, 0, ...)$ with a 1 in the n'th position is bounded but has no convergent subsequence. Therefore the Bolzano-Weierstrass property is a property that some, but not all, normed spaces have. We will talk about this property later.

**Definition.** Let $V$ be a normed space. A sequence $x_n$ in $V$ is said to be Cauchy if

$$(\forall \varepsilon > 0) \, (\exists N \in \mathbb{N}) \, (\forall n, m \geq N) \, \|x_n - x_m\| < \varepsilon$$

**Definition.** A normed space is said to be complete if all Cauchy sequences converge. Intuitively, this means it has no "gaps".

The fact that convergent sequences are Cauchy and Cauchy sequences are bounded holds with the same proof as in $\mathbb{R}$, since normed spaces satisfy the triangle inequality.

**Proposition.** If a sequence $(x_n)$ in a normed space is Cauchy and it has a convergent subsequence $(x_{n_k})$ then it is convergent.

*Proof.* Let $x$ be the limit of that subsequence. Fix $\varepsilon \to 0$. Then let k be so large that for $m > n \geq n_k$ we have that $\|x_n - x_m\| < \varepsilon$ and also $\|x_{n_k} - x\| < \varepsilon$. Now for each $m$ that is that large, by the triangle inequality, setting $n = n_k$,

$$\|x - x_m\| \leq \|x_n - x_m\| + \|x - x_n\| < 2\varepsilon$$

$\square$

From the definition we note that lipschitz equivalence preserves whether a sequence is Cauchy, and also preserves completeness. Note also that $\mathbb{R}^n$ is complete.

**Example.** Note that since a sequence of functions being uniformly Cauchy implies it is uniformly convergent, $C([0,1])$ with the supremum norm is complete. However, it is not complete with the $L^1$ norm: Consider, for example, the following sequence of functions converging to an indicator function of $\left[0, \frac{1}{2}\right]$ in $L^1$:

$$f_n(x) = \begin{cases} 1 & x \leq \frac{1}{2} \\ 1 - n\left(x - \frac{1}{2}\right) & \frac{1}{2} < x < \frac{1}{2} + \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

**Example.** Subspaces of complete spaces need not be complete: Consider $(0,1)$ in $\mathbb{R}$ and the sequence $\frac{1}{n}$.

Definitions and properties of open and closed sets in normed spaces are almost exactly the same as in $\mathbb{R}^n$.

**Definition.** Let $(V, \|\cdot\|)$ be a normed space. A subset $E \subseteq V$ is said to be open if for all $y \in E$ there is an $r > 0$ such that $B_r(y) \subseteq E$

**Definition.** Let $(V, \|\cdot\|)$ be a normed space. A subset $E \subseteq V$ is said to be closed if $V \setminus E$ is open

As in $\mathbb{R}^n$ we think of closed as meaning "contains all of its boundary" and open means "contains none of its boundary". Also, closed sets contain all their limit points, and sets that contain all their limit points are closed. Also, finite intersections and arbitrary unions of open sets are open; finite unions and arbitrary intersections of closed sets are closed.

**Definition.** Let $(V, \|\cdot\|)$ be a normed space and $E$ be any subset and let $F \subseteq E$. Then $F$ is said to be open in $E$ if for every $y \in F$, there is some $r > 0$ such that $(B_r(y) \cap E) \subseteq F$.

## 8.3 Compactness

Recall (level 6.2, 6.4) that if a subset of $\mathbb{R}^n$ is closed and bounded, then it has the following 2 properties:

1. Every sequence in the set has a subsequence converging to a point in the set

2. Every cover of the set by open sets has a finite subcover.

In general normed spaces, we call property 1 sequential compactness and property 2 compactness. We will show that in normed spaces, and in the more general notion of metric spaces, these are equivalent. You should keep in mind that they are indeed equivalent in metric spaces - we will prove it in the chapter on metric spaces. However, they are not equivalent in general topological spaces, which we will encounter in the relevant sublevel.

We want to show that in $\mathbb{R}^n$ both of these properties which we will later show are equivalent to eachother is eqiuvalent to being closed and bounded. It turns out that it is easy to show that property 1 implies being closed and bounded. To do this, note that if a set is unbounded then a sequence of increasingly large points that get larger without bound has no convergent subsequence, and if the set is missing a boundary point then a sequence in that set converging to that missing boundary point cannot converge to any point in the set since the point it would converge to is missing.

The statement that in $\mathbb{R}^n$, closed and bounded is equivalent to property 2, is known as the Heine-Borel theorem.

## 8.4 Continuous functions between normed spaces

Let $(V, \|\cdot\|)$ and $\left(V', \|\cdot\|'\right)$ be normed spaces and let $E \subseteq (V, \|\cdot\|)$ be a function. Then it is said to be continuous if

$$(\forall x \in E)\,(\forall \varepsilon > 0)\,(\exists \delta > 0)\,\|x - y\|_V < \delta \implies \|f(x) - f(y)\|_{V'} < \varepsilon$$

**Theorem.** Let $V$ be a normed space. A function defined on $E \subseteq V$is continuous if and only if pre-images of open sets are open in $V$.

*Proof.* Let $f$ be continuous and $W$ be an open subset of $V$, then we want to show $f^{-1}(W)$ is open. For each $\varepsilon > 0$ small enough such that W contains $B_\varepsilon(f(x))$ (at least one exists since W is open), there is $\delta > 0$ such that

$$|x - y| < \delta, y \in E \Rightarrow |f(x) - f(y)| < \varepsilon$$

so $f(y) \in B_\varepsilon(f(x)) \subseteq W$ if $y \in B_\delta(x) \cap E$. Since this is true for each $y$ in $B_\delta(x) \cap E$ we have that $f(B_\delta(x)) \cap E \subseteq W$. Therefore we have that

$$B_\delta(x) \cap E \subseteq f^{-1}(B_\varepsilon(f(x)))$$

and since$x$ was an arbitrary element in the pre-image of W, this means an open ball around $x$ is contained in $f^{-1}(B_\varepsilon(f(x)))$ which is in $f^{-1}(W)$, so $f^{-1}(W)$ is open in $E$ due to every arbitrary point having a set of the form $B_\delta(x) \cap E$ around it contained in $f^{-1}(W)$.

For the other direction, if this "open set" property holds, then if $f(x) = y$ then the pre-image of $(y - \varepsilon, y + \varepsilon)$ is open so every point in this pre-image has an open ball around it, including x, so there is a $\delta$ such that $(x - \delta, x + \delta)$ is in the pre-image.

□

Keep this alternative characterization of continuity in mind - it will be used as the definition of continuity when we study topological spaces.

With the same proof as in the normal continuous function case, a function $f$ is continuous at $y$ if and only if $y_k \to y \implies f(y_k) \to f(y)$.

**Theorem.** Let $(V, \|\cdot\|)$ and $\left(V', \|\cdot\|'\right)$ be normed spaces and let $K$ be a sequentially compact subset of $V$ and let $f : V \to V'$ be continuous. Then the image $f(K)$ is sequentially compact in $V'$, closed, and bounded.

Once we can prove this it will immediately imply that if $V' = \mathbb{R}$ then the function attains its supremum and infimum, since those are limit points of the closed image and they exist by boundedness.

*Proof.* Let $(x_k)$ be a sequence in $f(K)$ with $x_k = f(y_k)$ for some $y_k \in K$, then there is a subsequence $\left(y_{k_j}\right)$ such that $y_{k_j} \to y$ by sequential compactness of $K$, then $f\left(y_{k_j}\right) \to f(y)$ so $x_{k_j} \to f(y) \in F(K)$ so $F(K)$ is sequentially compact, hence closed and bounded by earlier discussion.

□

## 8.5 Equivalence of norms on finite dimensional spaces

**Lemma.** Let $V$ be an n-dimensional real vector space with a basis $\{v_1, v_2, ..., v_n\}$. Then for any $x \in V$ write $x = \sum x_j v_j$ for $x_j \in \mathbb{R}$ and $\|x\|_2 = \sqrt{\sum x_j^2}$. Then $S = \{x \in V : \|x\|_2 = 1\}$ is sequentially compact in $(V, \|\cdot\|_2)$

*Proof.* Given a sequence $x^{(k)}$ in $S$ write $x^{(k)} = \sum_{j=1}^n x_j^{(k)} v_j$. Then by Bolzano-Weierstrass in $\mathbb{R}^n$ there is a subsequence of this such that each coordinate $x_j$ converges. Therefore, since the norm is a continuous function in the coordinates, we have that for the limit of the subsequence $x_j^{(k_l)} = \tilde{x}_j$, we know $\sum_{j=1}^n \tilde{x}_j v_j \in S$ and the subsequence $x^{(k_l)}$ converges to that point.

□

**Theorem.** Any two norms on a finite dimensional real vector space are equivalent (and same for a complex vector space, by treating it as a real vector space)

*Proof.* We want to show that if $\|\cdot\|$ is any other norm then it is equivalent to $\|\cdot\|_2$. Given a basis $\{v_1, v_2, ..., v_n\}$ for $V$, write

$$x = \sum_{j=1}^n x_j v_j$$

$$\|x\| = \left\|\sum_{j=1}^n x_j v_j\right\| \leq \sum_{j=1}^n |x_j| \|v_j\| \leq \|x_2\| \sqrt{\sum_{j=1}^n \|v_j\|^2}$$

by Cauchy-Schwarz, so $\|x\| \leq b\|x\|_2$ for $b = \sqrt{\sum_{j=1}^n \|v_j\|^2}$.

Now consider $\|\cdot\| : (S, \|\cdot\|_2) \to \mathbb{R}$. We know that for any $x, y \in V$ we have $\|x - y\| \leq b\|x - y\|_2$. By the triangle inequality, we know that

$$\|x\| \leq \|x - y\| + \|y\| \text{ and } \|y\| \leq \|x - y\| + \|x\|$$

Therefore $|\|x\| - \|y\|| \leq \|x - y\| \leq b\|x - y\|_2$. This implies that when $x$ is close to $y$ under $\|\cdot\|_2$, $|\|x\| - \|y\||$ is small. Therefore the function $\|\cdot\| : (S, \|\cdot\|_2) \to \mathbb{R}$ is continuous, thus (by sequential compactness of $S$ and therefore its image under a continuous function) attains a non-zero lower bound $a$, so $a\|x - y\|_2 \leq \|x - y\|$

$\square$

This means properties such as being Cauchy or Convergent don't depend on the choicee of a norm in a finite dimensional space, so all bounded sequences have convergent subsequences and sequential compactness is equivalent to being closed and bounded and the space is complete, regardless of norm.

# 9 Metric spaces

## 9.1 Basic properties

This generalizes normed spaces.

**Definition.** Let $X$ be any set. A metric on $X$ is a function $d : X \times X \to \mathbb{R}_{\geq 0}$ such that

1. $d(x, y) = d(y, x)$

2. $d(x, y) = 0$ if and only if $x = y$

3. $d(x, y) \leq d(x, z) + d(z, y)$ (Triangle inequality)

The pair $(X, d)$ is called a metric space.

Note that all normed spaces are metric spaces with $d(x, y) = \|x, y\|$. Open balls, open and closed sets, continuity, convergence, Cauchy are defined the same way as in normed spaces.

**Example.** Given any set $X$ we can define the discrete metric by

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

**Example.** Given a metric space $(X, d)$ we can define $g(x, y) = \min(1, d(x, y))$ and $h(x, y) = \frac{d(x,y)}{1+d(x,y)}$. In both cases, we obtain a bounded metric.

**Proposition.** The triangle inequality holds for $g, h$ above so they are actually metrics.

*Proof.* Note that

$$g(x, y) = \min(1, d(x, y)) \leq \min(1, d(x, z) + d(z, y)) = \min(1, d(x, z) + d(z, y))$$

If $d(z, y) > 1$ then $g(x, z) + g(z, y) \geq 1 \geq g(x, y)$ so done. Otherwise, $d(z, y) = g(z, y)$ so we have after the above chain of inequalities,

$$\leq \min(1 - d(z, y), d(x, z)) + g(z, y) \leq g(x, z) + g(z, y)$$

so done for $g$.

For $h$, note that we start with $d(x,y) \leq d(x,z) + d(z,y)$ and we use the fact that $\frac{1}{1+\frac{1}{x}}$ is an increasing function for $x > 0$ to get that for $a, c \geq 0, b, d > 0$ we have

$$\frac{a}{b} \leq \frac{c}{d} \iff \frac{a}{a+b} \leq \frac{c}{c+d}$$

Then we have for $b = d = 1$ and $a = d(x,y)$ and $c = d(x,z) + d(z,y)$ that

$$\frac{d(x,y)}{1+d(x,y)} \leq \frac{d(x,z)+d(z,y)}{1+d(x,z)+d(z,y)} \leq \frac{d(x,z)}{1+d(x,z)} + \frac{d(z,y)}{1+d(z,y)}$$

as required.

□

**Definition.** Metrics are $d, d'$ Lipschitz equivalent if there exists $B > A > 0$ such that

$$Ad(x,y) \leq d'(x,y) \leq Bd(x,y)$$

Note that Lipschitz equivalent metrics give the same open sets, but the converse is not true. For example, if $d(x,y) = \min\{1, |x-y|\}$ on $\mathbb{R}$ then this is not Lipschitz equivalent to the standard metric despite giving the same set of open sets.

**Definition.** Given a metric space $X$ and a point $x \in X$, a neighbourhood of $x$ is a set containing an open set that contains $X$.

For the same reason as in $\mathbb{R}^n$, finite intersections and arbitrary unions of open sets are open; finite unions and arbitrary intersections of closed sets are closed.

**Proposition.** Let $(X,d)$ be a metric space, then $x_k \to x$ if and only if for every neighbourhood $V$ of $x$ there exists $K$ such that $x_k \in V$ for all $k \geq K$.

*Proof.* Note that since every open ball is a neighbourhood, if a sequence has this open set property it converges.

Conversely suppose $x_k \to x$. Let $V$ be any neighbourhood of $X$, then since $V$ contains an open set around $x$ it contains an open ball $B_\varepsilon(x)$ and by definition of convergence the $x_k$ are eventually always inside this open ball.

□

**Definition.** A sequence is bounded if it is entirely contained within some open ball.

Note that in a metric space, bounded sequences are contained in an open ball about any point, since $B_r(x) \subseteq B_{r+d(x,y)}(y)$ - easy to check by the triangle inequality.

## 9.2 Cauchy sequences and completeness

Properties of Cauchy sequences (Specifically, that convergent sequences are Cauchy, Cauchy sequences are bounded and Cauchy sequences with convergent subsequences converge) hold with the same proof as in the normed space case.

**Definition.** A metric space is complete if Cauchy sequences converge.

**Example.** Define $h : \mathbb{R}^n \to \mathbb{R}^n$ by $h(x) = \frac{x}{1+\|x\|}$ where $\|x\|$ is the standard Euclidean/Pythagoras norm. Then this is injective since if $h(x) = h(y)$ it is easy to see that the magnitude and direction of both $x$ and $y$ must coincide. To check that $d(x,y) = \|h(x) - h(y)\|$ is a metric, note that $h$ bijects $\mathbb{R}^n$ into $B_1(0)$ and then $d$ acts like the standard metric on the open ball under this bijection. But then $B_1(0)$ is incomplete so $\mathbb{R}^n$ is incomplete with this metric. This means that when we lose the additional structure of a normed space, we lose the idea that $\mathbb{R}^n$ is complete with any metric.

**Theorem.** Let $(X,d)$ be a metric space and let $Y \subseteq X$ be any subset. Then

1. If $(Y, d|_{Y\times Y})$ is complete then $Y$ is closed in $X$.

2. If $(X, d)$ is complete then $Y$ is closed in $X$ if and only if $(Y, d|_{Y\times Y})$ is complete.

*Proof.* 1. Let $x \in X$ be a limit point of $Y$, then there is some sequence $x_k \to x$ where each $x_k \in Y$. Since $(x_k)$ is convergent, it is a Cauchy sequence so it is Cauchy in $Y$, so it must converge to some point in $Y$ which must be $x$, so $X$ contains all its limit points and is therefore closed.

2. The complete implies closed direction is what we just proved. On the other hand if $Y$ is closed and $(x_k)$ is a Cauchy sequence in $Y$ then it is Cauchy in $X$ and its limit point in $X$ (exists by completeness of $X$) is in $Y$ so $Y$ is complete.

□

## 9.3 Compactness

Finally at last we prove the big theorem I've promised.

**Lemma.** (Lebesgue number lemma) let $(X, d)$ be a sequentially compact metric space and let $\mathcal{U}$ be an open cover, then there exists a $\delta > 0$ such that each ball $B_\delta(x)$ is inside some $U_x \in \mathcal{U}$.

*Proof.* Suppose for a contradiction no such $\delta > 0$ exists. Then for each $n = 1, 2, 3, ...$ there is an $x_n$ such that $B_{\frac{1}{n}}(x_n)$ is not contained in any element of $\mathcal{U}$. By sequential compactness $x_n$ has a convergent subsequence, say $x_{n_i} \to x$. Let $x \in U \in \mathcal{U}$ and let $\varepsilon > 0$ be such that $B_\varepsilon(x) \in U$. Then for $i$ large enough we have $x_{n_i} \in B_{\frac{\varepsilon}{2}}(x)$ and $\frac{1}{n_i} < \frac{\varepsilon}{2}$. By the triangle inequality we then have $B_{\frac{1}{n_i}}(x_{n_i}) \subseteq B_\varepsilon(x_\infty) \subseteq U \in \mathcal{U}$ which is a contradiction.

□

**Theorem.** A metric space $(X, d)$ (or a subset of a metric space interpreted as a metric space) is compact if and only if it is sequentially compact

*Proof.* **Step 1.** Assume that it is not sequentially compact.

Then there is some $(t_n)$ in $X$ with no convergent subsequence, so for each $x \in X$ there is an open ball around it with only finitely many $t_i$'s, call this $U_x$. Then $\mathcal{U} = \{U_x : x \in X\}$ is an open cover of $X$ and cannot have a finite subcover as such a subcover would contain the whole sequence $(t_n)$ but we know it only contains a finite amount of the sequence.

**Step 2.** Assume that it is sequentially compact.

Given an open cover let $\delta > 0$ be a Lebesgue number for this cover by the lemma above.

**Step 2.1.** We will show that there exists a finite set $A \subseteq X$ such that $X = \bigcup_{a\in A} B_\delta(a)$.

If this is not the case, then for every finite set $A \subseteq X$ there exists an $x \in X$ with $d(x, a) \geq \delta$ for all $a \in A$. If this were the case then we could construct a sequence $x_1, x_2, ...$ in $X$ by choosing $x_i$ to have a distance $\geq \delta$ from all previous elements of the sequence and this would have no convergent subsequence contradicting compactness.

**Step 2.2.** We will show that this implies the result.

To do this note for each $a \in X$ that $B_\delta(a)$ lies inside some $U_a \in \mathcal{U}$. Then $\{U_a : a \in A\} \subseteq \mathcal{U}$ is a finite subcover by step 2.1 so $(X, d)$ is compact.

□

Note that all compact spaces are bounded - this is easy to see directly from both definitions (an unbounded set has both a sequence with no convergent subsequence and an open cover with no finite subcover).

**Corollary.** All compact spaces are complete since Cauchy sequences have convergent subsequences and are therefore convergent.

**Definition.** (Totally Bounded) A metric space $(X, d)$ is said to be totally bounded if for all $\varepsilon > 0$ there is a finite covering of $X$ by open balls of radius $\varepsilon$. Clearly, totally bounded implies bounded.

**Theorem.** Let $(X, d)$ be a metric space. Then $X$ is compact if and only if $X$ is complete and totally bounded.

*Proof.* **Step 1.** Let $X$ be complete and totally bounded.

Let $y_i$ be a sequence in $X$. For every $j \in \mathbb{N}$ there exists a finite set of points $E_j$ such that every point is within $\frac{1}{j}$ of a point in $E_j$ by total boundedness. Since $E_1$ is finite there must exist $x_1 \in E_1$ such that there are infinitely many $y_i$'s in $B_1(x_1)$, call the first $y_i$ in this open ball (exists by the well ordering principle) $y_{i_1}$.

Now there is some $x_2 \in E_2$ such that there are infinitely many $y_i$'s in $B_1(x_1) \cap B_{\frac{1}{2}}(x_2)$. Pick the smallest such value of $i > i_1$ and call this $y_{i_2}$. Keep doing this until infinity.

Doing this gives a Cauchy subsequence (Since $m > n$ implies $d(y_{i_m}, y_{i_n}) < \frac{2}{n}$), and this converges by completeness of $X$.

**Step 2.** Let $X$ be compact, then it is complete as discussed above so we just need to show that it is totally bounded. This is clear since for all $\varepsilon > 0$ we can cover it with (possibly infinitely many) balls of radius $\varepsilon$ then take a finite subcover by compactness.

□

## 9.4 Continuous functions

**Definition.** A continuous function between metric spaces is defined the same way as in normed spaces and with the same proof a function is continuous if and only if pre-images of open sets are open.

**Definition.** Let $f$ be a function between metric spaces with domain being a metric space $X$ and codomain being a metric space $Y$. Then $f$ is uniformly continuous if

$$(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x, y \in X)\, d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon$$

With the same proof as in the real case, a continuous function on a compact metric space is uniformly continuous.

**Definition.** A function $f$ defined on a metric space $X$ is said to be Lipschitz continuous if there exists a finite constant $K$ such that $d_Y(f(x), f(y)) \leq K d(x, y)$. Note that it is easy to see that

$$\text{Lipschitz continuous} \implies \text{Uniformly continuous} \implies \text{Continuous}$$

**Example.** Uniformly continuous does not imply Lipschitz continuous: Consider $\sqrt{x}$ on $[0, 1]$.

Note that two metrics $d, d'$ being Lipschitz equivalent is equivalent to both the identity maps $i : (X, d) \to (X, d')$ and $i' : (X, d) \to (X, d')$ being Lipschitz continuous.

Note that the metric *itself* (by picking a pivot point that we are measuring the distance from) is Lipschitz continuous with constant 1. Explicitly, if $x$ is the pivot point, then $d(x, z) - d(x, y) \leq d(y, z)$ by the triangle inequality

## 9.5 The contraction mapping theorem (To be completed)

# 10 Multivariate analysis (To be completed)